

Chapter 8. Pressure-Support Ventilation

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Pressure-Support Ventilation: Introduction

Pressure-support ventilation (PSV) is a mode of partial ventilator support. Such modes are widely used in intensive care units (ICUs) because most ventilated patients (unless deeply sedated) have preserved respiratory drive. The use of these modes helps to reduce need for sedation, an important issue in the ICU,^{1,2} and potentially prevents disuse atrophy of the respiratory muscles that can result from controlled ventilation.^{3,4} This preventive effect has been shown experimentally with different modes of partial support.^{3,5} Finally, partial support may facilitate both the screening process for detecting patients able to breathe spontaneously as well as the weaning of patients with prolonged or difficult weaning.⁶ An ideal mode of partial support should be able to supply both full ventilator support and optimal support during weaning; optimize patient-ventilator synchronization and comfort while reducing the need for sedation and the risk of cardiovascular consequences; and, if possible, facilitate or reduce the duration of the weaning. PSV meets several of these requirements, at least partially, as discussed in this chapter. PSV also has limitations, which are delineated. One important limitation is that overassistance of the patient can be easily reached and improvement in the delivery of the optimal PSV level continues as a field of research.

PSV can be remarkably effective in reducing patient effort and avoiding respiratory distress, and can offer a comfortable ventilator support to many patients. PSV can also deliver support much in excess of patient needs and results in excessive delivered volume, excessive duration of inspiration relative to neural inspiratory time (T_I), or both. Much recent research has been undertaken to understand and analyze the consequences of delivering of excessive pressure. Many benefits of PSV, which provides greater freedom to the patient than traditional modes, can be obscured by improper usage.

Epidemiology

Some clinicians view PSV primarily as a mode devoted to weaning and only consider its use until late in a patient's course.⁶ An international survey on mechanical ventilation in 361 ICUs in twenty countries was conducted in 1998 (published in 2002).⁷ On the first day, PSV was used in less than 10% of all patients; the combination of SIMV with PSV was used in almost 15%, and assist-control in approximately 60% of the patients. A low level of PSV was used to perform a once-daily weaning attempt in 28% of such attempts, a gradual reduction of PSV was used as the sole weaning method in 21% of cases, and a gradual reduction of synchronized intermittent mandatory ventilation (SIMV) and PSV was used in 22% of all cases. Overall, PSV was used (one way or another) for 45% of weaning attempts, suggesting that clinicians consider weaning as the main indication for PSV.

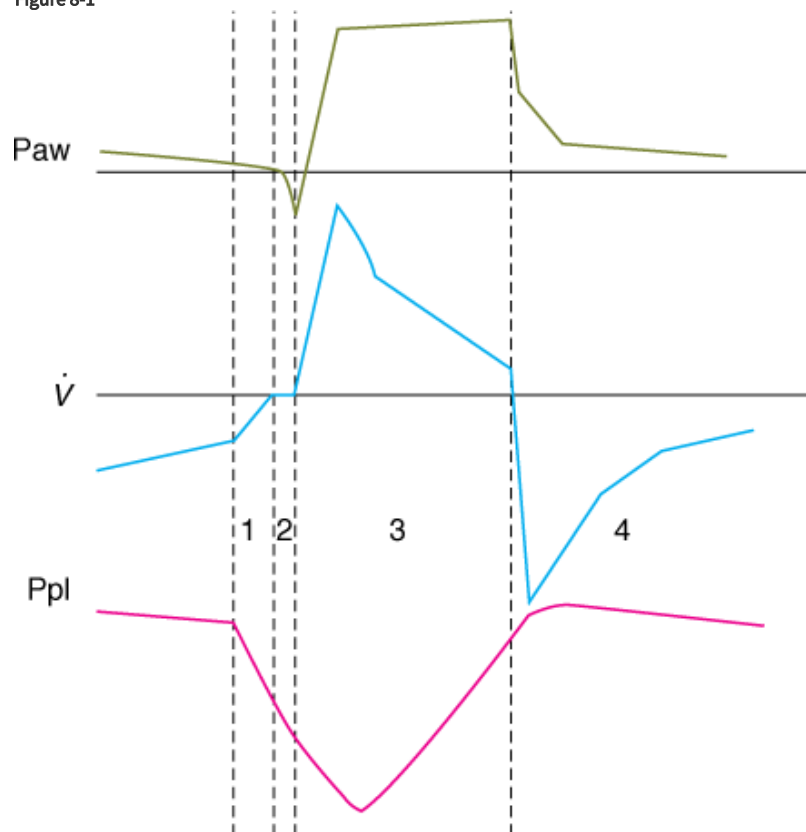
In 2004, Esteban et al repeated the prospective international observational cohort study, employing a nested comparative study performed in 349 intensive care units in twenty-three countries, and compared the findings with the

1998 cohort.⁸ Whereas the use of a T piece was the most common initial method for spontaneous breathing trials, the use of low levels of pressure support for weaning trended upward over time (10% vs. 14%). Among patients not extubated, methods for gradual withdrawal differed: there was a decrease in the use of SIMV and of SIMV combined with PSV, and a major increase in the use of PSV for weaning (19% vs. 55%) between the two observation periods. This international survey was repeated a third time, on a larger scale in 2010 and included more than 8000 patients. The initial results indicate that after more than 6 days of mechanical ventilation, PSV was now the most frequently used ventilator mode, indicating progressive dissemination in the use of this technique over the years.⁹

Differences among Mechanical Ventilators Definition and Phases

PSV is a pressure-targeted (or limited) mode in which each breath is patient-triggered and supported.^{10–13} It provides breath-by-breath support by means of a positive-pressure boost synchronized with inspiratory effort: patient initiated and flow terminated (Fig. 8-1). During inspiration, airway pressure is raised to the preset pressure-support level. The speed of pressurization is system specific but most recent ventilators offer the possibility of adjusting this pressurization rate. Throughout the inspiratory phase, the ventilator works as a pressurized demand-flow system at a predetermined pressure level. PSV is maintained until the machine determines the end of expiration, supposedly reflecting the end of patient demand. The expiratory trigger mechanism is based on decay of inspiratory flow. When inspiratory flow falls below a threshold value, which should indirectly indicate that the inspiratory muscles have relaxed, the ventilator cycles to the expiratory phase releasing the PSV and opening its expiratory port. A level of positive end-expiratory pressure (PEEP) lower than the inspiratory plateau pressure can then be applied. PSV can thus be defined as a patient-initiated (pressure or flow), pressure-targeted, flow-cycled mode of mechanical ventilation.

Figure 8-1



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A pressure-supported breath with tracings of airway pressure (P_{aw}), flow ($\dot{V}$), and pleural pressure (P_{pl}). Four phases of patient effort can be discerned. Phase 1 is still expiratory and corresponds to an effort performed against intrinsic positive end-expiratory pressure; it occurs before the triggering system of the ventilator can detect any signal that indicates the onset of patient inspiratory effort. Phase 2 is the time required to activate the triggering system of the ventilator (also called the initiation phase). Phase 3 is the insufflation phase during which the ventilator pressurizes the airway at the level set by the clinician. This phase is terminated by the cycling-off criterion. Patient inspiratory effort may terminate before the end of this phase. Phase 4 is the expiratory phase.

Three phases of PSV can be distinguished: (a) recognition of the beginning of inspiration, (b) pressurization, and (c) recognition of the end of inspiration. These phases constitute the working principles of PSV, and can vary from one ventilator to another (see [Fig. 8-1](#)). As discussed in the section [Differences Among Mechanical Ventilators](#), these variations may induce differences in the effect of PSV for similar levels of pressurization.

Initiation of the Cycle

Triggering of inspiration is initiated by patient effort and is detected by a pressure or flow sensor. Trigger sensitivity is adjustable. This mechanism requires an active effort by the patient, the intensity of which depends on the characteristics of the valve. The opening time delay varies between 50 and 250 milliseconds, depending on the ventilator.^{14–18} The most recent data indicate that most ventilators now respond in less than 100 milliseconds.^{19–22}

Opening of the demand valve can be triggered by a fall in pressure or a difference in the flow signal between inspiratory and expiratory flows (referred to as *flow-by*). For the latter, a constant flow is delivered to the circuit during the expiratory phase; inspiratory effort is then detected as a small difference between inspiratory and expiratory flow. Flow-triggering avoids the need for a closed demand valve. Aslanian et al²³ showed that the difference between pressure-triggered and flow-triggered systems has become quite small on modern ventilators. The triggering phase represents less than 10% of a patient's overall effort to breathe. A flow-triggering system makes a statistically significant difference but of limited clinical importance.

Pressurization

Once inspiration has been initiated, the ventilator delivers a high inspiratory flow, which rapidly decreases throughout the rest of inspiration. A servo regulatory mechanism maintains the proper flow to reach the appropriate preset PSV level and keeps this pressure approximately constant until expiration occurs. Flow regulation varies among ventilators, thus determining the pressure waveform. Usually, the servo valve is continuously controlled during the breath, such that delivered pressure closely approximates target pressure set by the clinician. In general, the aperture of the proportional servo valve is progressively reduced as the monitored pressure gets closer to the target pressure. For this reason, the wave shape often constitutes a pressure ramp rather than a true square wave. The pressure level can be adjusted between 0 (spontaneous breathing through the ventilator circuit) and a maximum of 30 or 60 cm H₂O (even more with some ventilators). In clinical settings, pressure levels above 30 cm H₂O are rarely used.

Pressure increases according to a rate that is system-specific; formerly, it was nonadjustable. A high speed of pressurization produces a square pressure wave; low achievement of the preset PSV level attenuates this shape. Many ventilators now allow adjustment of the rate of pressurization. Its influence is discussed in the section [Pressurization Rate and Inspiratory Flow](#).

Cycling of Expiration

During PSV, cycling to exhalation is primarily triggered by a decrease of inspiratory flow from its peak to a system-specific threshold value. This critical decrease of inspiratory flow is taken as indirect evidence that the inspiratory muscles have begun to relax. Expiration is triggered when either an absolute level of flow (between 2 and 6 L/min) or a fixed percentage of peak inspiratory flow (12% or 25%) is reached, depending on the ventilator model ([Table 8-1](#)). The threshold value for cycling, which can be viewed as sensitivity of the expiratory trigger, was formerly nonadjustable. Adjustment is now offered to clinicians on many ventilators.

Table 8-1: Technical Characteristics of Pressure Support Ventilation and Available Settings on Intensive Care Ventilators

		Inspiratory Trigger		Pressurization	Cycling-off Criterion		
	Manufacturer	Flow	Pressure		Flow Cycle	Pressure Cycle	Time Cycle
PB 7200	Puritan Bennett/ Covidien	1 to 15	0.5 to 20		5 L/min	+1.5 cm H ₂ O	3 s
PB 740	Puritan Bennett/ Covidien	1 to 20			10 L/min or 25% PF	+3 cm H ₂ O	3.5 s
PB 760	Puritan Bennett/ Covidien	1 to 20		5% to 100%	Adjust. 1% to 45% PF	+3 cm H ₂ O	3.5 s
PB 840	Puritan Bennett/ Covidien	1 to 20		5% to 100%	Adjust. 1% to 80% PF	+1.5 cm H ₂ O	3 s
Evita 2	Drager				25% PF	High pressure limit	
Evita 2 dura	Drager	0.3 to 15 L/min		0 to 2 s	25% PF	High pressure limit	
Evita 4	Drager	0.3 to 15 L/min		0 to 2 s	25% PF	High pressure limit	4 s
Evita XL	Drager	0.3 to 15 L/min		0 to 2 s	5% to 70% PF	High pressure limit	
Savina	Drager				25% PF	High pressure limit	
Servo 900C	Maquet		0 to 20		25% PF	+3 cm H ₂ O	
Servo 300	Maquet	0.6 to 2	0 to 20		5% PF	+20 cm H ₂ O	

		Inspiratory Trigger		Pressurization	Cycling-off Criterion		
	Manufacturer	Flow	Pressure		Flow Cycle	Pressure Cycle	Time Cycle
Servo-i	Maquet	0.6 to 2	0 to 20	0 to 0.4 s	Adjust. 1% to 80% PF	High pressure limit	
Servo-s	Maquet	0.6 to 2	0 to 20	0 to 0.4 s	Adjust. 1% to 80% PF	High pressure limit	
Veolar	Hamilton				25% PF	High pressure limit	3 s
Galileo	Hamilton	0.5 to 15	0.5 to 10	25 to 200 ms	Adjust. 10% to 40% PF	High pressure limit	3 s
Raphael	Hamilton						
Bird 8400	Viasys Healthcare	1 to 10	1 to 20		25% PF	High pressure limit	3 s
T-Bird	Viasys Healthcare	1 to 20			5% to 30% PF (submenu)	High pressure limit	3 s
Vela	Viasys Healthcare	1 to 8			Adjust. 5% to 30% PF	High pressure limit	0.3 to 3 s
Avea	Viasys Healthcare	0.1 to 20	0.1 to 20		Adjust. 5% to 45% PF	High pressure limit	0.15 to 5 s
Bear 1000	Viasys Healthcare				25% PF		5 s
Bipap Vision	Respironics/Philips	Automatic (autotrack)		0.05 to 0.4 s	Automatic (autotrak)		

		Inspiratory Trigger		Pressurization	Cycling-off Criterion		
	Manufacturer	Flow	Pressure		Flow Cycle	Pressure Cycle	Time Cycle
Esprit	Respironics/Philips	0.5 to 20	0 to 20	0.1 to 0.9 s	Adjust. 10% to 45% PF		
LTV1000	Pulmonetics	1 to 9			Adjust. 10% to 40% PF	High pressure limit	
Elisee	Saime				Adjust. 10% to 40% PF or automatic		
e500	Newport				Adjust. 5% to 50% PF, variable	High pressure limit	
HT50	Newport		0 to -10			High pressure limit	0.1 to 3 s
Infrasonics star	Infrasonics		-0.5 to 20		4 L/min or 10% PF		3.5 s
Inspiration	Event	1 to 25	-1 to 20	Fast/medium/low	Adjust. 10% to 80%		

Abbreviations: *Adjust.*, adjustable; *PF*, peak-flow.

Detection of a small degree of pressure (1 to 3 cm H₂O) above the fixed PSV level, consequent to sudden expiratory effort by the patient, can also be used (alone or combined with the flow criteria) to stop inspiratory assistance. Finally, a time limit for inspiration is usually included. This serves as a safety mechanism if a leak develops in the circuit and the two previous methods of terminating inspiration become inoperative. Complications have been reported in the absence of this time-limit mechanism, whereby constant insufflation (at the PSV level) creates a high level of continuous positive airway pressure.²⁴

Other Settings

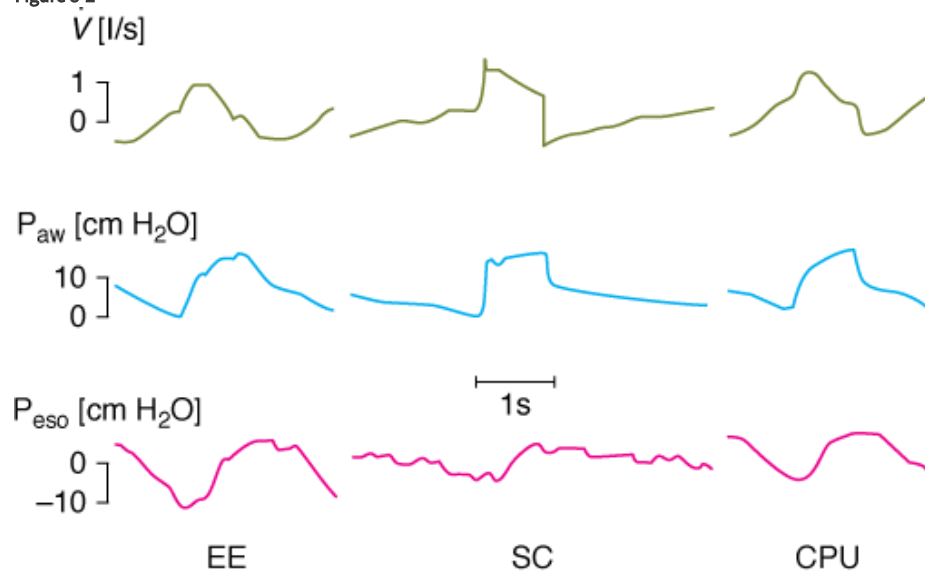
Because no mandatory breath is present with PSV, a safety feature is often available in case of apnea. This may be an automatic feature, or a minimal frequency, or minute ventilation to be set. The time delay for apnea may be adjustable. This safety feature is not available on all ventilators.

PSV can be used in conjunction with SIMV.^{25–28} Two approaches have been used: addition of a fixed level of PSV during spontaneous breathing to overcome endotracheal tube (ETT) or circuit resistance,²⁹ or use of a variable level of PSV between the mandatory breaths. The second approach introduces considerable complexity into ventilator management of patients.

Differences among Mechanical Ventilators

During PSV, specific characteristics of the ventilator may interfere with patient respiratory activity. These differences may be determined by the manufacturer's algorithm to deliver pressure, such as speed of pressurization and/or initial peak flow setting, ability to maintain a plateau pressure and quality of regulation, and termination criteria used to cycle from inspiration to expiration. Nonspecific features include characteristics of the demand valve and/or triggering mechanism, and flow-impeding properties of the expiratory circuits, including PEEP devices.²² These differences may also vary with the type of ventilator, whether it is designed only for delivery of noninvasive PSV or a full intensive care ventilator.^{20,21,30,31} The relative weight of each factor is difficult to determine and may vary from one patient to another. This consideration should, however, be kept in mind when interpreting the results of clinical studies of PSV using various ventilators. One study with old-generation ICU ventilators compared three of them and found major work differences,³² showing that different characteristics of PSV could have a major influence on its efficacy (Fig. 8-2). Fortunately, most recent ventilators have designed much better systems of regulation, providing more homogeneous delivery of PSV, but clinically relevant differences still exist.²²

Figure 8-2



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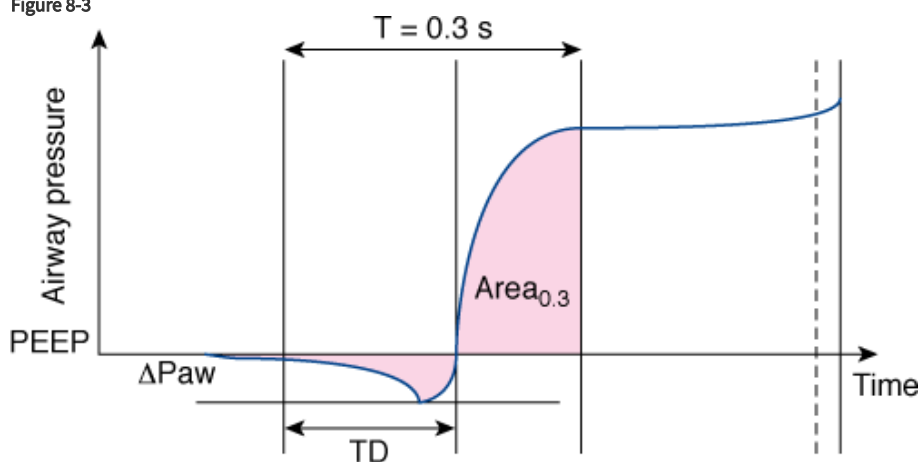
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The influence of a ventilator and its specific algorithm on patient effort. Three ventilators were studied at 15 cm H₂O of PSV. Flow, airway pressure (P_{aw}) and esophageal pressure (P_{eso}) are presented. Note the different airway and flow profiles, and the impact on esophageal pressure swing. Work of breathing was significantly less with SC. Although modern ventilators tend to homogenize the delivery of PSV, differences still exist and illustrate the effects of varying the pressure ramp. CPU, CPU1, Ohmeda, Maurepas, France; EE, Erica Engstrom, Bromma, Sweden; and SC, Servo 900 C, Siemens, Lund, Sweden. Reproduced from.²⁸

As with other assisted modes, the triggering mechanism is a key determinant of the efficacy of PSV. A poorly functioning demand valve has two consequences: it imposes an effort to open the valve, and it prolongs the time before assistance is delivered. Assisted modes, like PSV, are primarily devoted to reducing or optimizing this effort. Demand valves function with an unalterable delay before delivering gas flow to the patient. For instance, if 200 milliseconds is required between the beginning of an inspiratory effort and the opening of the valve, nearly one-third of the duration of inspiratory effort in a tachypneic patient may take place without any gas entering the lungs. If auto (or intrinsic) PEEP is present, another 200 milliseconds may be wasted (while the respiratory muscles work against this positive alveolar pressure) before any inspiratory flow can start.³³ In addition, if the speed of pressurization of PSV is low, another 200 milliseconds is required to reach the plateau pressure. Thus, assistance will be delivered to the patient 600 milliseconds after the beginning of inspiratory effort, which may correspond to the end of that patient's inspiratory effort.³⁴

Comparison of the triggering functions of various ventilators demonstrates that the most recent generation of ventilators, using pressure-sensitive mechanisms, flow-sensitive mechanisms, or both, usually require less effort and open faster than the older generation. This was extensively studied by Richard et al,¹⁹ and subsequently by Thille et al²² who compared different generations of ventilators, including the new turbine ventilators (Figs. 8-3, 8-4, 8-5). The ability of different ventilators to pressurize the airway during PSV was investigated. Different levels of simulated inspiratory demand were used. Pressurization was assessed through the net area of the inspiratory airway pressure-time tracing over the first 0.3 second, 0.5 second, and 1 second at different levels of PSV (Figs. 8-3 and 8-4). Triggering sensitivity was assessed independently by measuring the time delay and the pressure fall with different levels of inspiratory drive (Figs. 8-3 and 8-5). Ventilators released after 1993 achieved significantly better results than most previous generation ventilators regarding the pressure-time area at 0.3 seconds and triggering delay, indicating large improvements in terms of triggering and pressurization. Regarding PSV and trigger performances, this generation of ventilators outperformed most previous generation ventilators; this was also the case for some piston and turbine-based ventilators, including several of those specially designed for noninvasive ventilation (NIV). Six years later, trigger function, pressurization capacity and accuracy of pressure measurements during simulated PSV, and expiratory resistance were again evaluated in a similar bench study.²² In 2006, new-generation turbine-based ventilators performed as well as, or better than, the best compressed-gas ventilators. The newest ventilators did not perform significantly better than the 2000 ventilators, suggesting that a technological ceiling may have been reached.

Figure 8-3

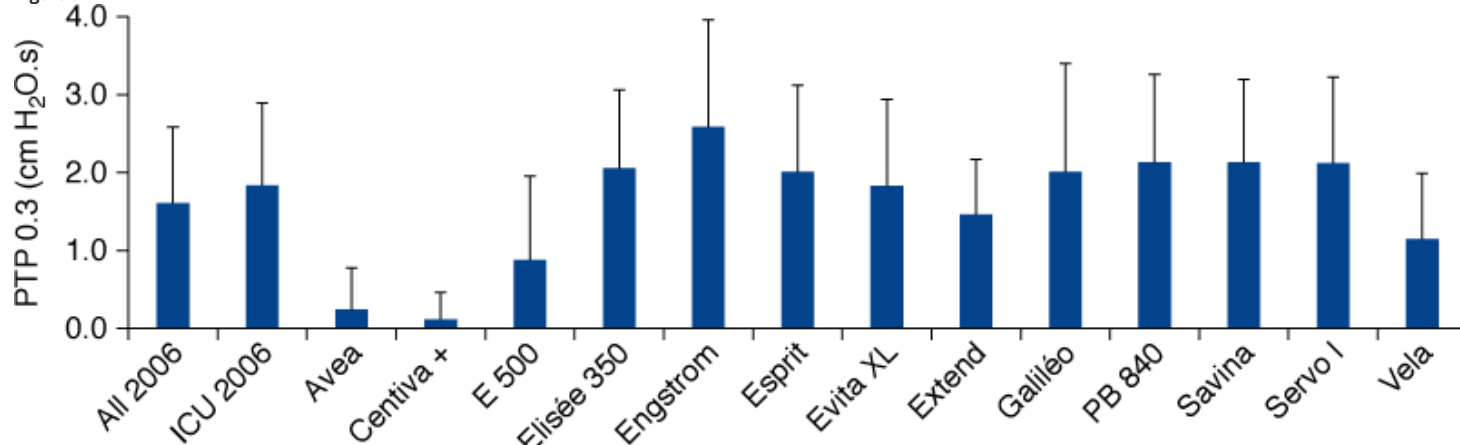


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The method used to calculate the trigger characteristics and pressurization phase during pressure support ventilation based on the airway pressure-time curve. The total trigger phase is evaluated by the time delay (TD) between the onset

of simulated effort and the time at which airway pressure becomes positive after experiencing a pressure fall (ΔP_{aw}). The quality of pressurization is best quantified as the area measured at 0.3 second. (With kind permission from Springer Science and Business Media: Cox D, Tinloi SF, Farrimond JG. Investigation of the spontaneous modes of breathing of different ventilators. *Intensive Care Med.* 1988;14:532–537.)

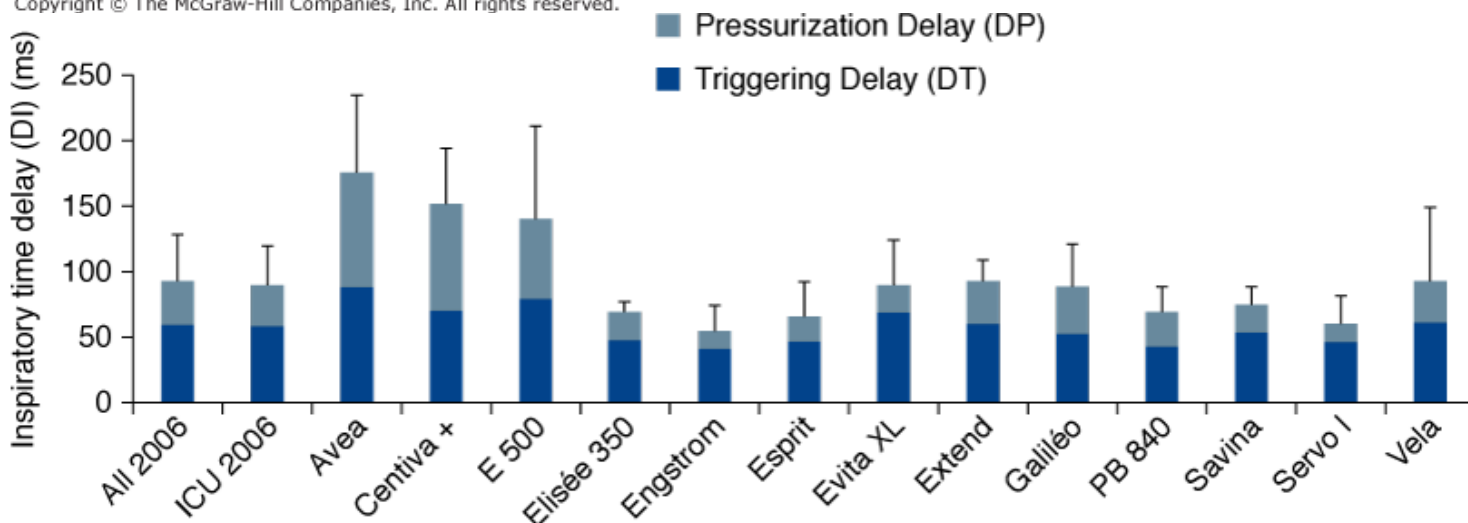
Figure 8-4



A

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B

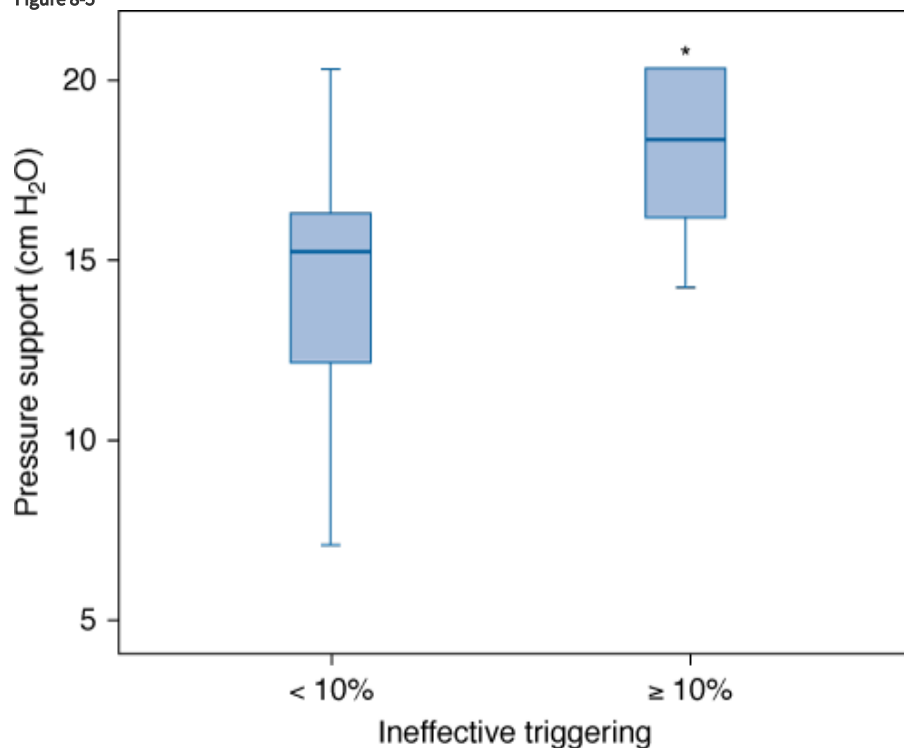
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A. Inspiratory area, measured as the integral of the airway pressure-time trace over the first 0.3 second of inspiration (see legend of Fig. 8-3), for a PSV level of 15 cm H₂O for the same simulated level of inspiratory demand. **B.** Inspiratory time delay in milliseconds measured during the same tests, separated into triggering and pressurization delay. The new-generation ICU ventilators proposed in 2006 were evaluated. These included Avea and Vela (Viasys Healthcare, Conshohocken, PA), E 500 (Newport Medical Instruments, Costa Mesa, CA), Elisée 350 (Resmed-Saime, North Ryde, Australia), Engstrom and Centiva (General Electric, Fairfield, CT), Esprit (Respironics, Murrysville, PA), Extend (Taema, Antony, France), Savina and Evita XL (Dräger, Lubeck, Germany), Galileo (Hamilton, Rhäzuns, Switzerland), PB 840 (TYCO,

Carlsbad, CA), and Servo I (Maquet, Solna, Sweden). Of these thirteen ventilators, four were turbine-based (Elisée 350, Esprit, Savina, Vela) and nine were conventional servo-valve compressed-gas ventilators. Seven were viewed as ICU ventilators (Avea, Evita XL, Engstrom Extend, Galileo, PB 840, Servo I) and six were mid-level ICU ventilators (Centiva, E 500, Elisee 350, Esprit, Savina, and Vela). The mean values for all ventilators or only ICU ventilator is also presented. (With kind permission from Springer Science and Business Media: Thille AW, Lyazidi A, Richard JC, Galia F, Brochard L. A bench study of intensive-care-unit ventilators: new versus old and turbine-based versus compressed gas-based ventilators. *Intensive Care Med.* 2009;35:1368–1376.)

Figure 8-5



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Relationship between the level of PSV and the frequency of ineffective triggering in a cohort series of sixty-two consecutive patients requiring mechanical ventilation for more than 24 hours. *Box plots* show median, interquartile range (25th to 75th percentiles), and outliers (5th to 95th percentiles) of PSV in patients with and without a high prevalence of ineffective triggering (>10%). PSV was higher in patients with a high incidence of ineffective triggering. * $p < 0.05$. (With kind permission from Springer Science and Business Media: Thille AW, Rodriguez P, Cabello B, Lellouche F, Brochard L. Patient-ventilator asynchrony during assisted mechanical ventilation. *Intensive Care Med.* 2006;32:1515–1522.)

Dedicated Noninvasive Ventilators

NIV using PSV can be performed with either turbine ventilators, specially conceived to provide NIV, or conventional ICU ventilators originally designed for invasive ventilation. The rate of use of each of these is variable depending on the care setting, the etiology of respiratory failure, the specialist, and even the geographic region.^{35–37} Ventilators delivering PSV and PEEP, termed *bilevel ventilation*, and designed for home ventilation have been evaluated in stable, awake patients with chronic ventilatory failure.³⁸ Despite some variability in the delivery of pressure, no difference was found in terms of comfort or improvement in inspiratory muscle unloading. These differences, however, might have greater impact in patients with acute respiratory failure.

The presence of air leaks around the mask is a major problem related to use of NIV. When ICU ventilators are not capable of compensating for leakage, leaks generate autotriggering, prolonged cycling, ineffective efforts, and a poor pressurization capacity of the ventilator. For these reasons, manufacturers have developed specific algorithms, called *NIV modes*, which aim to minimize the negative impact of leaks on ICU ventilator performance. Dedicated NIV ventilators are designed to function with leaks and have developed sophisticated algorithms that evaluate online the magnitude of a leak and readjust the triggering thresholds. Several bench studies show variable performance in the ability to ameliorate these key ventilator functions with the activation of the NIV mode,^{39,40} and an overall better performance for dedicated NIV ventilators. The clinical effect of “NIV modes” on patient–ventilator interactions has also been evaluated in two recent short-term clinical studies. Vignaux et al compared four ICU ventilators, with and without the activation of the NIV mode, showing a reduction in patient–ventilator asynchronies with the NIV mode.⁴¹ A second study, employing the same schema but also testing a specific NIV ventilator, showed a significant reduction in the rate of these events with the latter.⁴² Although clinical studies are needed to clarify this issue, it is advisable to use NIV modes on ICU ventilators or dedicated ventilators.

Main Physiologic Effects of Pressure-Support Ventilation

Breathing Pattern

During PSV, the patient maintains control over respiratory rate, and has partial control of T_I and tidal volume (V_T). As such, PSV seems to allow the patient to breathe in a “physiologic” way. This is only partially true, because there is a complex interaction between ventilator support and patient control of breathing. This interaction depends on the pressure level and PSV characteristics. For instance, a change in the criterion for cycling from inspiration to expiration will result in a different T_I , different V_T , and may result in more (or less) dynamic hyperinflation.

The addition of PSV modifies the spontaneous breathing pattern.^{43–48} Most patients develop an increase in V_T and decrease in respiratory rate with increasing levels of PSV. The breathing pattern adapts rapidly under PSV when the respiratory muscles face a new workload.⁴⁹ Adjustment of the PSV level can be guided by noting the breathing pattern response: changes in breathing pattern in response to loading conditions usually occur within 1 to 2 minutes.^{49,50} Importantly, evaluation of a patient during a low level of PSV is different from evaluation of during a T-tube trial, for instance, even if the clinical outcome of the two approaches may be similar. Two studies show that the use of a low level of PSV during a “spontaneous breathing trial” modifies the thresholds that may be used to predict weaning success or weaning failure.^{51,52}

Excessive levels of support, however, can generate hyperinflation, respiratory alkalosis, respiratory depression with apnea, or appearance of missing efforts.⁵³ A high level of support-induced hyperinflation may indeed result in an inability to trigger the ventilator (so-called ineffective triggering), with a substantial difference in the ventilator’s displayed respiratory rate and the patient’s true respiratory rate.⁵⁴ The frequent occurrence of these findings (discussed below in [Degree of patient-ventilator synchrony or asynchrony](#)) illustrates the fact that the insufflation during PSV is often terminated after cessation of the patient’s inspiratory time, especially in case of airway obstruction.⁵⁵ The higher the pressure level, the longer is this prolongation.⁵⁶ Thus, to some extent, PSV artificially forces the patient to decrease respiratory rate as a way of trying to maintain a sufficient expiratory time.⁵⁴ As the PSV level is increased, an imbalance

almost inevitably occurs between prolongation of the insufflation time and shortening of expiratory time, promoting severe asynchrony.

In patients with acute lung injury, factors other than respiratory muscle load influence respiratory drive. Pesenti et al reported that variation in arterial oxygen (O_2) saturation between 85% to 90% and 100%, obtained by modifying fractional inspired oxygen concentration (FI_{O_2}), had a significant effect on respiratory drive in patients with acute lung injury receiving PSV.⁵⁷ Volta et al reported similar findings: modulation of produced variation in dyspnea, occlusion pressure, and respiratory frequency.⁵⁸

The influence of PSV on the duty cycle (fractional inspiratory time, T_I/T_{TOT}) is variable and influenced by the setting of the pressure ramp on the ventilator.⁵⁹ A decreasing duty cycle with increasing PSV levels was observed in several studies.^{44–46}

The influence of PSV on minute ventilation is variable, producing an increase or no change.^{43–47} An increase in minute ventilation is often observed when PSV is compared with unassisted breathing through the ventilator circuit. More frequently, an increase in PSV fails to substantially modify minute ventilation whereas it modifies alveolar ventilation. Consequently, the breathing pattern may be markedly modified without significant change in minute ventilation.^{43–48} Thus, monitoring minute ventilation is of little help when titrating the level of PSV.

Gas Exchange and Distribution of Ventilation and Perfusion

The primary goal of PSV is to support patient effort while allowing a satisfactory gas exchange. PSV is not primarily aimed at improving oxygenation. The effects of PSV on gas exchange are primarily explained by increased alveolar ventilation resulting from changes in breathing pattern. Indeed, despite lack of change in minute ventilation, an increase in V_T produces a decrease in the ratio of dead space to tidal volume (V_D/V_T). Thus, alveolar ventilation is often increased. Other factors may influence arterial blood gases, such as changes in O_2 consumption, modification of total dead space, and altered distribution of ventilation. During the weaning of patients with hypercapnic respiratory failure, addition of PSV produced a correction of the partial pressure of arterial carbon dioxide (Pa_{CO_2}) and respiratory acidosis compared to spontaneous breathing.⁴³ In normal, nonintubated subjects, PSV 10 cm H_2O produces significant decreases Pa_{CO_2} in below normal levels.⁵³ PSV can thus correct hypoventilation but also induce hyperventilation that is not counteracted by respiratory motor output.⁶⁰ Consequently, although PSV permits correction of hypercapnia resulting from rapid shallow breathing or helps patients with chronic CO_2 retention to choose their own target Pa_{CO_2} , the level of PSV requires fine adjustment as respiratory alkalosis can easily occur.

Although breathing pattern, especially respiratory frequency, is in part controlled by the patient, an interaction exists between the level of PSV and alveolar ventilation, which is not fully controlled by respiratory center command.

MacIntyre and Leatherman showed in a lung model that a biphasic effect can occur with increasing levels of PSV.⁶¹ Above a certain limit, passive (hyper-) inflation will result with high levels of PSV. That excessive assistance with PSV may induce respiratory alkalosis not controlled by the patient's respiratory centers and has important consequences. Parthasarathy and Tobin found that during sleep PSV was associated with numerous episodes of apneas, desaturation, and microarousals leading to sleep fragmentation.⁶² This could be prevented by adding dead space to the circuit. Not tested in this study was the likelihood that reductions in the level of PSV might prevent this problem.

The distribution of ventilation and perfusion during PSV has been assessed in a few studies.^{63–68} Valentine et al⁶³ compared SIMV, PSV, and airway pressure release ventilation in nine patients a few hours following cardiac surgery. The major characteristics of ventilation–perfusion ($\dot{V}_A/\dot{Q}$) distributions were similar with all modes.⁶³ Dead space was lower during airway pressure release ventilation than during either SIMV or PSV. Gas exchange was assessed with the six-inert-gas technique in a study comparing controlled mechanical ventilation, unassisted spontaneous breathing, and PSV 10 cm H₂O.⁶⁴ Using isotopic scanning, they evaluated regional distribution of $\dot{V}_A/\dot{Q}$ ratios in eight patients with chronic obstructive pulmonary disease (COPD). Ventilator discontinuation was associated with rapid shallow breathing and an increase in perfusion to low $\dot{V}_A/\dot{Q}$ regions. Isotopic scans revealed a horizontal craniocaudal difference of $\dot{V}_A/\dot{Q}$ with all modes, and the lowest $\dot{V}_A/\dot{Q}$ ratios were found at the bases. Abnormalities in $\dot{V}_A/\dot{Q}$ distribution observed during spontaneous breathing were also present during 10 cm H₂O PSV, but to a smaller extent.

Ferrer et al assessed whether PSV could improve $\dot{V}_A/\dot{Q}$ imbalance during the transition between positive-pressure ventilation and spontaneous breathing in seven intubated patients with COPD during weaning.⁶⁶ PSV avoided $\dot{V}_A/\dot{Q}$ worsening during this transition. Hemodynamics, blood gases, and $\dot{V}_A/\dot{Q}$ distributions were equivalent during PSV and assist-control ventilation (ACV) when the two modes provided similar levels of assistance. Diaz et al studied the reasons for improvement in partial pressure of oxygen (P_{O_2}) and partial pressure of carbon dioxide (P_{CO_2}) in ten patients with acute hypercapnic exacerbations of COPD who were switched from spontaneous breathing to PSV during NIV.⁶⁵ Improvement in blood gases was primarily mediated by a higher alveolar ventilation, and not improvement in $\dot{V}_A/\dot{Q}$ relationships. Although O₂ uptake tended to decrease, the respiratory exchange ratio increased, explaining a slight increase in arterial-to-alveolar O₂ difference secondary to increased clearance of body stores of CO₂ during NIV. These results suggested that attaining an efficient breathing pattern rather than high inspiratory pressures should be the primary goal for improving arterial blood gases during NIV with PSV in this type of patient. Lastly, a recent study assessed gas exchange and changes in lung volume through the use of lung diffusion capacity for carbon monoxide (DL_{CO}) in sixteen patients without COPD.⁶⁹ An increase in PSV of 5 cm H₂O neither affected lung volume nor increased the volume of the lung participating in gas exchange, but was associated with a slight but significant deterioration in DL_{CO}. Thus, a target V_T closer to 6 mL/kg than to 8 mL/kg of predicted body weight during PSV was associated with better gas exchange.

The effect of PSV on oxygenation varies and depends on many factors, such as the induced changes in alveolar ventilation, O₂ consumption, dead space, and mean airway pressure.^{43–46} Most investigators have not found significant changes in arterial oxygenation when PSV was compared with other modalities (primarily spontaneous breathing or SIMV) delivered at the same FiO₂. Compared with continuous positive pressure ventilation in surgical ICU patients, Zeravik et al suggested that only patients with a low level of extravascular lung water had improved oxygenation with PSV among patients with moderate acute respiratory failure.⁷⁰

Work of Breathing and Respiratory Effort

A major goal of PSV is to assist respiratory muscle activity in a way that improves the efficacy of patient effort and decreases workload. Many of the initial studies on PSV have focused on this point and have measured work of breathing or indexes of patient effort during PSV.^{11,13,43,45,46,71,72}

MacIntyre was one of the first to study the effects of various levels of PSV in patients.¹¹ The level of PSV was positively correlated with V_T and negatively correlated with respiratory rate. He suggested that PSV alters the characteristics of work of breathing: the change in the pressure-to-volume ratio of the work of each breath decreased progressively with increasing levels of PSV. In intubated patients recovering from acute respiratory failure, Brochard et al compared breathing characteristics during 10 cm H₂O PSV, spontaneous unassisted breathing through a ventilator, and a continuous flow system without a demand valve.¹³ PSV produced significant increases in V_T and partial pressure of arterial oxygen (Pa_{O_2}), and a decrease in respiratory rate, transdiaphragmatic pressure (Pdi) swings, pressure-time index, and electromyographic activity of the diaphragm. Subsequently, Brochard et al compared several levels of PSV in eight patients who were experiencing weaning difficulties, four of whom had COPD.⁴³ During unassisted breathing, patients breathed with a small V_T and a high rate, a pattern associated with unsuccessful weaning, hypoxemia, and hypercapnia. All patients exhibited intense activity of their sternocleidomastoid muscles and the analysis of the diaphragmatic electromyographic recordings suggested impending high-frequency fatigue during PSV0. All these signs or symptoms disappeared at 10 cm H₂O or 20 cm H₂O of PSV, while activity of the sternocleidomastoid muscles was minimized or no longer present. Work of breathing returned to normal, whereas the respiratory rate remained around 30 breaths/min. These findings were later confirmed by another study also analyzing an index of high-frequency fatigue.⁷³

The values of respiratory rate described above emphasize that trying to “normalize” respiratory rate much below the point where a patient is no longer in respiratory distress, such as targeting a threshold of 20 breaths/min or even lower values, may not be desirable; we will see later that it may favor asynchrony. A limit of 30 breaths/min was also found by Jubran et al to be predictive of a inspiratory pressure-time product of less than 125 cm H₂O.sec/min, representing a desirable level of inspiratory effort.⁷⁴ This has important clinical implications for the bedside titration of PSV.

PSV acts with great efficiency in decreasing work of breathing. This is more or less proportional to the level of PSV and is accompanied by changes in breathing pattern measurable at the bedside, together with changes in respiratory muscle recruitment. There is, however, an individual limit of pressure above which work of breathing is not decreased and the patient becomes to be overassisted.⁴⁴

Different indexes have been used to assess respiratory muscle activity. Beck et al compared the crural diaphragmatic electrical activity (Edi) with Pdi during varying levels of PSV in intubated patients.⁵⁶ Changes in PSV did not alter neuromechanical coupling of the diaphragm: Edi and Pdi decreased proportionally with the addition of PSV. In contrast, Fauroux et al found that diaphragmatic pressure-time product, often used to quantify loading and unloading of the diaphragm, did not exhibit a linear relationship with the diaphragmatic electromyographic activity during PSV and that flow measurements may be necessary when assessing diaphragmatic unloading during PSV.⁷⁵

Compensation for the Work Caused by Endotracheal Tube and Demand Valve

PSV has been used to predict patient tolerance of unassisted breathing and extubation.⁷⁶ The idea is based on selecting a level of PSV just sufficient to overcome the circuit resistance. Thus, spontaneous muscular activity should be similar to what a patient would perform in the absence of an ETT or circuit.⁷⁷ The pressure needed to obtain a “adequate” breathing pattern during a spontaneous breathing test can provide insight into a patient’s ability to tolerate extubation.

It has long been argued that breathing through an ETT and demand valve increases respiratory muscle work⁷⁸ and that PSV can compensate for this increased demand.^{77,79,80} Part of the confusion, however, comes from the fact that the resistance posed by an ETT is probably close to upper-airway resistance after extubation. Several clinical studies have compared work of breathing before and immediately after extubation.^{80–82} These studies demonstrated that the work of breathing was similar or even often higher after extubation than before extubation (while breathing through an ETT). This indicates that there is no rationale for compensating for the ETT in itself. What needs to be compensated for, however, is the ventilator circuit through which the patient is breathing, including the triggering system.

In intubated subjects breathing with various levels of PSV who were disconnected from the ventilator and finally extubated,⁷⁷ the level of PSV that compensated for extra work of breathing through the ETT and ventilator circuit was calculated post hoc. In patients with underlying lung disease, the PSV level that compensated for the additional work ranged from 8 to 14 cm H₂O, while it averaged 5 cm H₂O in patients free of lung disease. Based on various studies,^{83,84} it has been argued that a PSV level of 5 to 10 cm H₂O be provided when a patient is breathing through a demand valve.

Despite wide individual variation among patients regarding the best pressure to apply in physiologic studies, a simplified approach, based on the same principle, has been applied in several large clinical trials. These studies showed that a low level of PSV (7 to 10 cm H₂O) is overall as efficient as a T-piece trial in testing whether a patient can be separated from the ventilator and eventually extubated, despite the lack of individual titration of PSV.^{8,85–87}

Some studies suggested that this low PSV test may be easier to tolerate than a T-piece trial,⁸⁸ and measurements of muscular effort support this impression in cardiac patients.⁸⁹ Therefore, a T-piece trial seems to constitute a more challenging test than the low PSV test, at least in cardiac patients.

Effect of Instrumental Dead Space

Instrumental dead space is usually constituted by the flex-tube connector, the Y piece, and the humidification system. Heat and moisture exchangers and heated humidifiers constitute a resistive load,⁹⁰ but heat and moisture exchangers also add instrumental dead space, because they are positioned between the Y piece and the ETT. The mechanical characteristics of heat and moisture exchangers can substantially modify breathing pattern, effort to breathe, and gas exchange during PSV. These effects were assessed in several studies during PSV and invasive ventilation,^{90–95} and also studies during NIV,^{96,97} with consistent results. Adding dead space with the heat and moisture exchanger reproduced well-described effects of addition of CO₂ on breathing pattern. In intubated patients during PSV, for instance, Pelosi et al reported that work of breathing increased from 8.8 ± 9.4 J/min with a heated humidifier to 14.5 ± 10.3 J/min with an heat and moisture exchanger.⁹² These investigators suggested that increasing the level of PSV by 5 cm H₂O may be necessary to compensate for the increased work of breathing caused by heat and moisture exchangers dead space. Recently, it was shown, however, that small heat and moisture exchangers have a much lower dead space, and that their impact is therefore limited or negligible.⁹⁸

Sleep

Mechanical ventilation in the ICU is associated with an abnormal sleep pattern characterized by abnormal circadian distribution of sleep and numerous arousals, similar to the pattern found in sleep apnea patients.^{99,100} The exact

influence of mechanical ventilation on sleep fragmentation in ICU patients remains poorly understood, but the ventilator mode and its settings, as well as patient–ventilator interactions, can influence the degree of fragmentation and the quality of sleep.^{62,99,101–103}

The effects of the ventilator mode and settings were investigated in ICU patients by Parthasarathy and Tobin.⁶² Sleep fragmentation was increased with the use of PSV as compared with ACV, mostly because of central apneas caused by the level of PSV. The main mechanism of apnea was a decrease in P_{CO_2} , which could be avoided by adding an external dead space. This illustrated an undesirable effect of PSV, resulting from hyperventilation. During ACV, by contrast, the patient can trigger some or all breaths, and a minimal respiratory rate and the V_T are preset, and this backup setting protects against apnea. Conceivably, however, a similar effect could have been achieved by selecting lower levels of PSV. This important study thus raised relevant clinical questions for the selection of ventilator settings.

The hypothesis that the possible deleterious effect—hyperventilation—could be avoided by a more appropriate level of PSV was confirmed in subsequent studies, both in home-ventilated patients and in ICU patients. A study of outpatients with neuromuscular disease compared two settings for NIV with PSV, one based on clinical parameters and the other based on physiologic requirements as estimated by an assessment of the patient's respiratory effort and mechanics.¹⁰³ The physiologic setting was associated with improvements in sleep quantity and quality, as well as with a lower level of PSV. An association was found between reductions of ineffective efforts (a marker of patient–ventilator asynchrony, mainly related to hyperinflation) and a higher proportion of rapid eye movement (REM) sleep, a sleep period with extreme physiologic importance. The smaller number of ineffective efforts with the physiologic settings were ascribed to the lower intrinsic PEEP, possibly related to the lower V_T . The apnea index was also lower at the physiologic setting.

The effects of three ventilatory modes on sleep were compared in nonsedated ICU patients.¹⁰² The three modes were ACV, clinically adjusted PSV, and automatically adjusted PSV. With automatically adjusted PSV, the pressure level is adjusted in real-time based on V_T , respiratory rate, and end-tidal CO_2 , as described later in this chapter (Closed loop delivery).¹⁰⁴ The goal of automatically adjusted PSV is to adjust the level of PSV to the patient's ventilatory demand so as to avoid underassistance or overassistance. Sleep was severely altered in the fifteen patients, who exhibited reductions in REM sleep and marked sleep fragmentation. No differences, however, were found in sleep architecture, sleep efficiency, or sleep fragmentation between the three ventilator modes. The absence of significant differences may reflect an adequate adjustment of ventilator settings, as minute ventilation was similar with the three modes. In this study,¹⁰² central apneas and ineffective efforts were relatively uncommon and were similar with the three modes. These results suggest that the mode may be less important than adjustment of the settings of each mode.

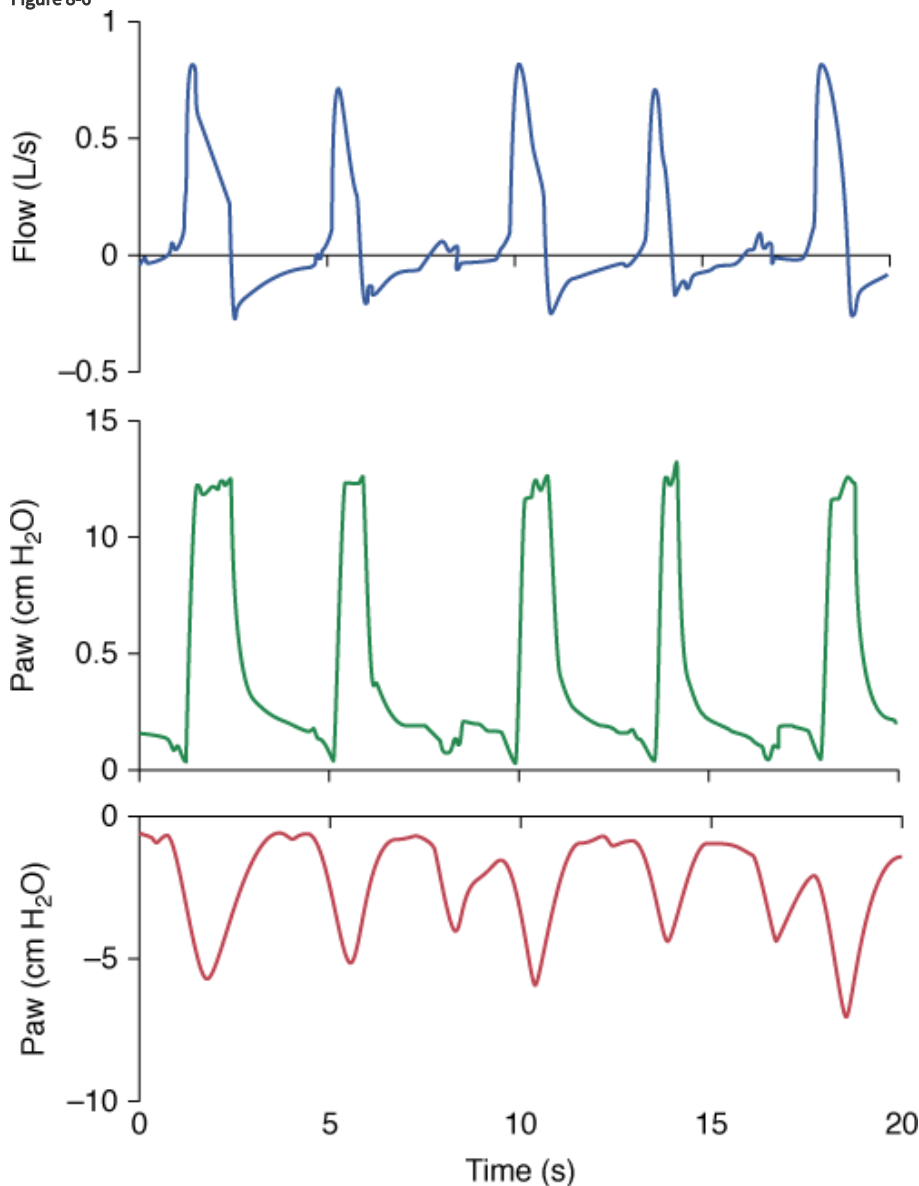
Newer modes may provide a more continuous and physiologic assistance. The effects of two assist-ventilation modes on sleep quality have been compared.¹⁰¹ Patients received proportional-assist ventilation (PAV) during one night and PSV during a second night. PAV involves applying a level of pressure that is proportional to the patient's inspiratory effort; therefore, the level of pressure is variable. The patient triggers all the breaths and V_T changes with every breath, reflecting the natural breath-to-breath variability. PSV, in contrast, applies the same level of pressure independently of the magnitude of a patient's inspiratory effort. The working hypothesis was that PAV would improve patient–ventilator interactions and lessen asynchrony, and therefore reduce sleep fragmentation. The ventilator was set to achieve the same reduction in inspiratory effort. The number of patient–ventilator asynchronies was significantly smaller with PAV than with PSV. In addition, PAV was associated with a lower fragmentation index and with higher percentages of slow-wave sleep and REM sleep. V_T and minute ventilation were lower with PAV, suggesting that this mode ensured better

matching of the assistance to the patient's requirements than did PSV, thereby decreasing the number of asynchronies linked to overassistance.

Degree of Patient–Ventilator Synchrony or Asynchrony during Pressure-Support Ventilation

The fundamental principle of assisted ventilation is to deliver assistance on a breath-by-breath basis in synchrony with patient effort. As discussed, some patient–ventilator asynchrony often exists with most current assisted modes, which can be aggravated by inappropriate settings, chiefly excessive support. Synchrony has been the subject of several investigations, often not specific to PSV.^{28,105–109} Patient–ventilator asynchrony has been described during invasive ventilation^{28,109–112} and NIV¹¹³ (Figs. 8-6, 8-7, 8-8, 8-9, 8-10, 8-11, 8-12, 8-13, 8-14).

Figure 8-6

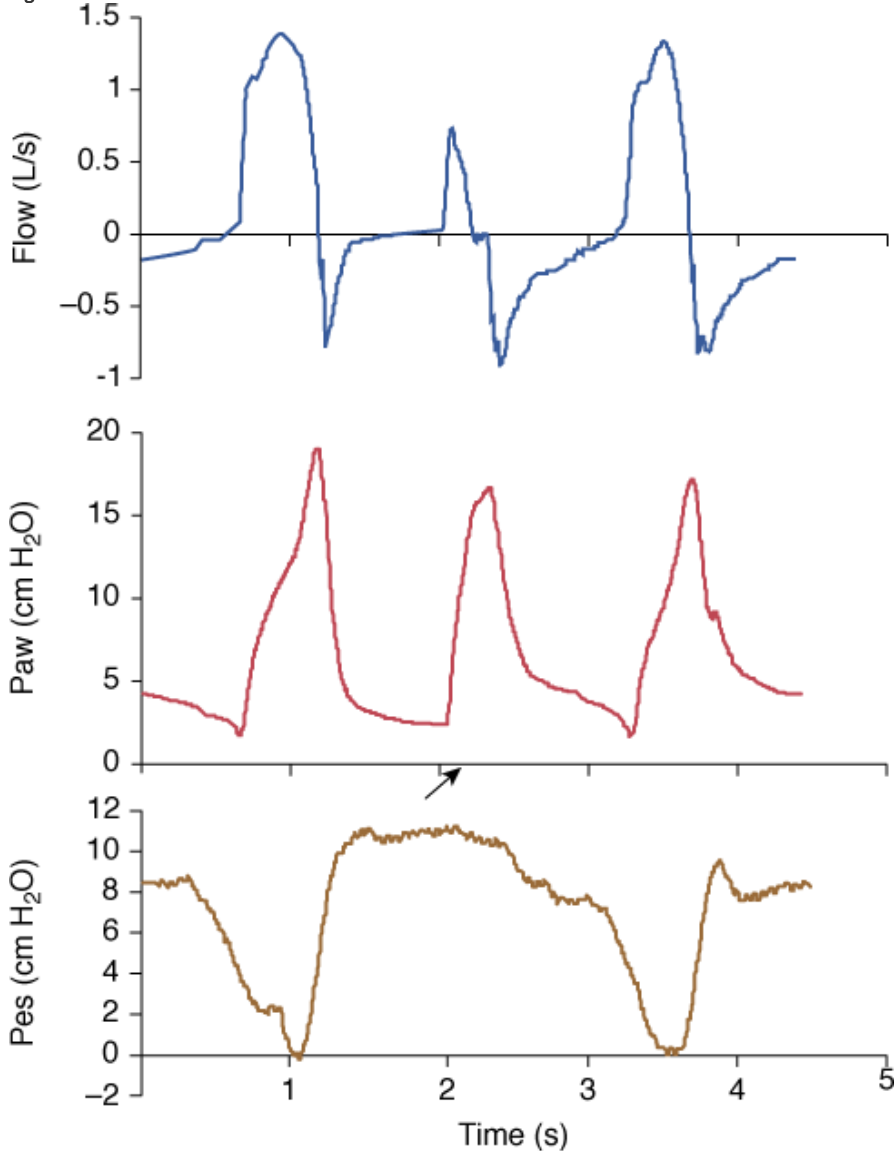


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Ineffective efforts. The third and sixth inspiratory efforts by the patient fail to trigger the ventilator. The efforts by the patient (visible on the esophageal pressure tracing [*Pes*]) are not accompanied by ventilator insufflations. A small and

transient increase in flow during expiration and a decrease in airway pressure are visible at the time of the failure-to-trigger events.

Figure 8-7

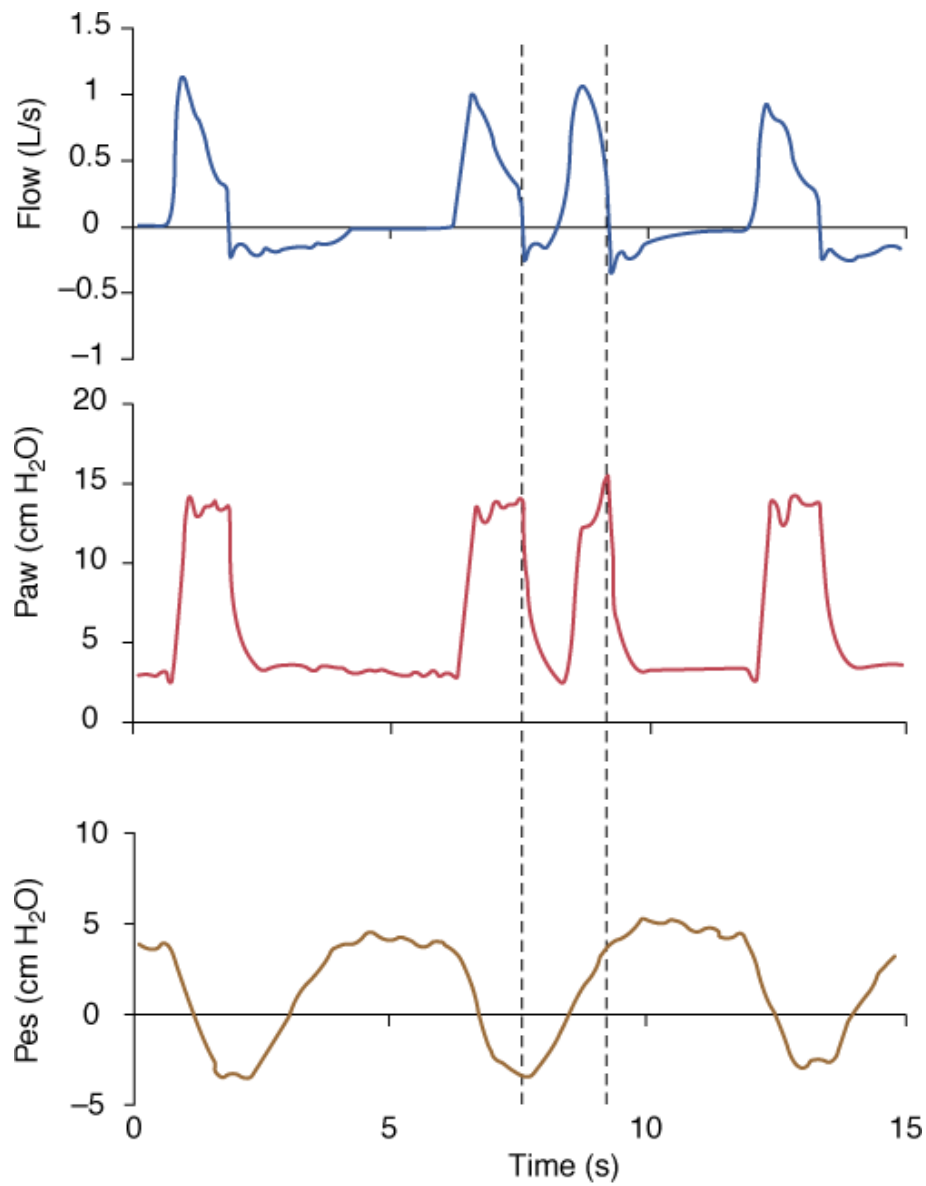


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Autotriggering. This form of asynchrony can occur when the inspiratory trigger is set too sensitive or in the presence of end-expiratory leaks. As on this tracing, a “short cycle” is a frequent result of an autotriggered cycle. The “autotriggered cycle” is accompanied by the absence of an initial airway pressure decay.

Figure 8-8

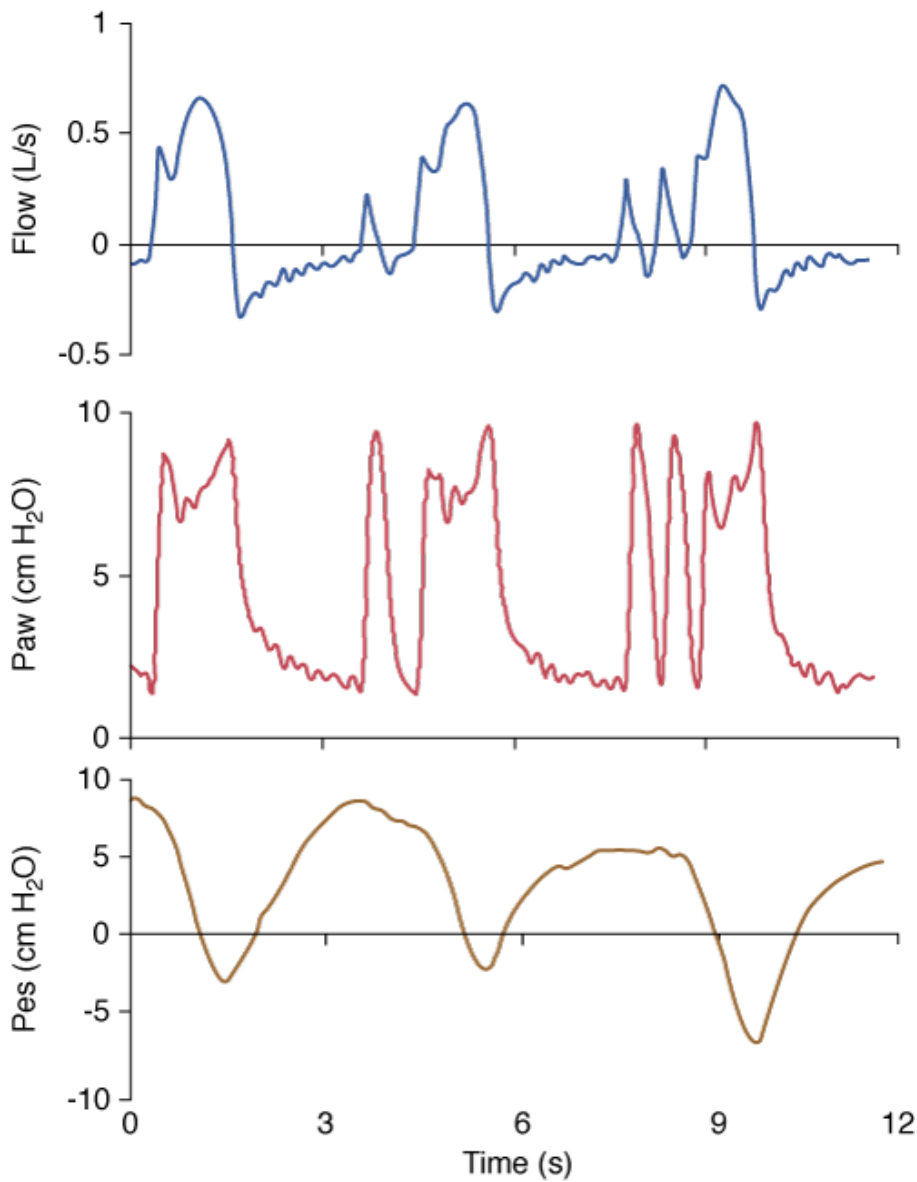


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Double cycles. Two ventilator cycles occur within a single patient's inspiratory effort. Three mechanisms can induce this asynchrony: autotriggering, high pressurization rate (present here with initial overshoot), and early cycling-off (also present in this case).

Figure 8-9

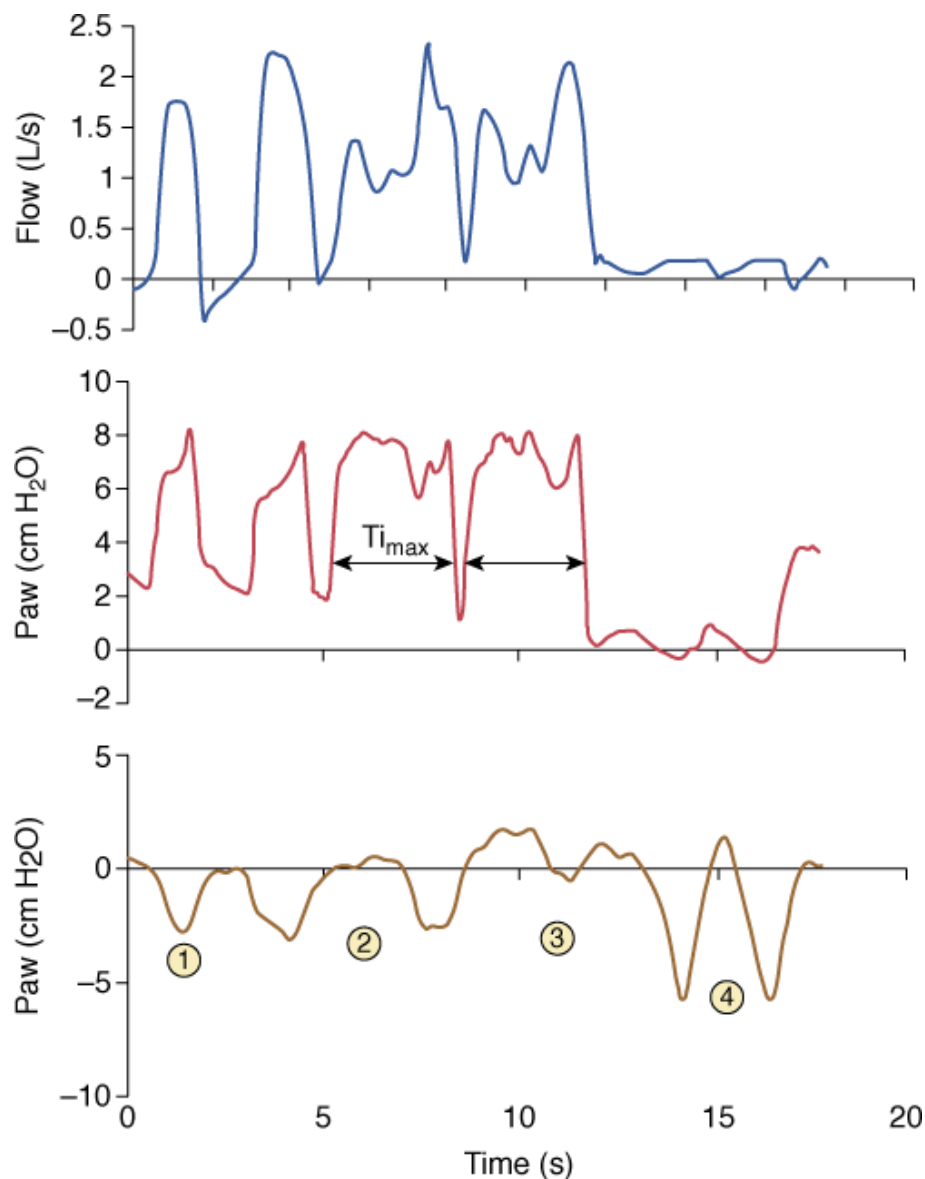


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Multiple cycles. Multiple cycles are frequently associated with autotriggered cycles and frequent “short cycles.” A high pressurization rate can favor this form of asynchrony. In this example, both autotriggering and high pressurization rate are present.

Figure 8-10

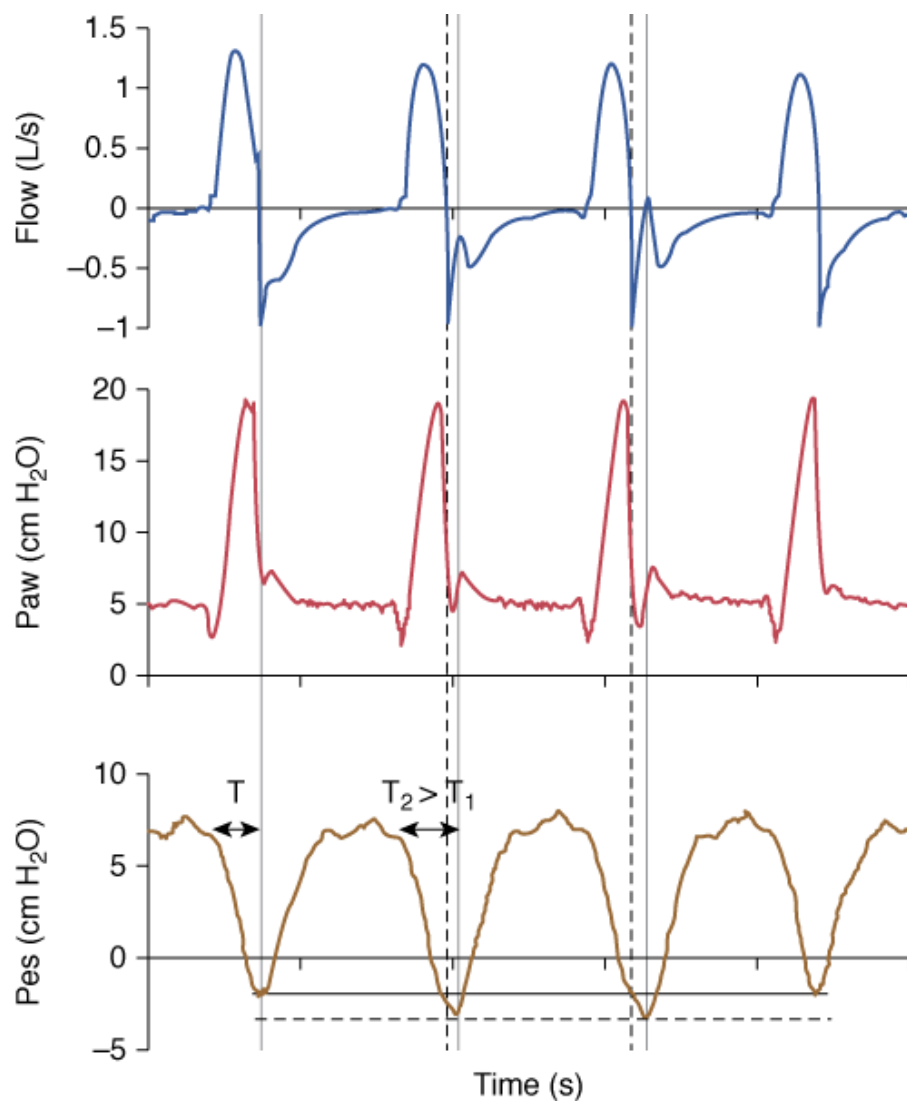


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Prolonged inspiration during noninvasive ventilation. This form of asynchrony during PSV results from a failure to recognize the flow cycling-off criterion. With an end-inspiratory leak, as in this example, the ventilator increases and/or sustains flow to maintain the set airway pressure (here above 2 L/s). This prevents recognition of the decelerating flow threshold and cycling to expiration. Insufflation is stopped only when maximum inspiratory time ($T_{i_{max}}$) is reached. Ineffective triggering secondary to hyperinflation may follow the prolonged inspiration, as in this example.

Figure 8-11

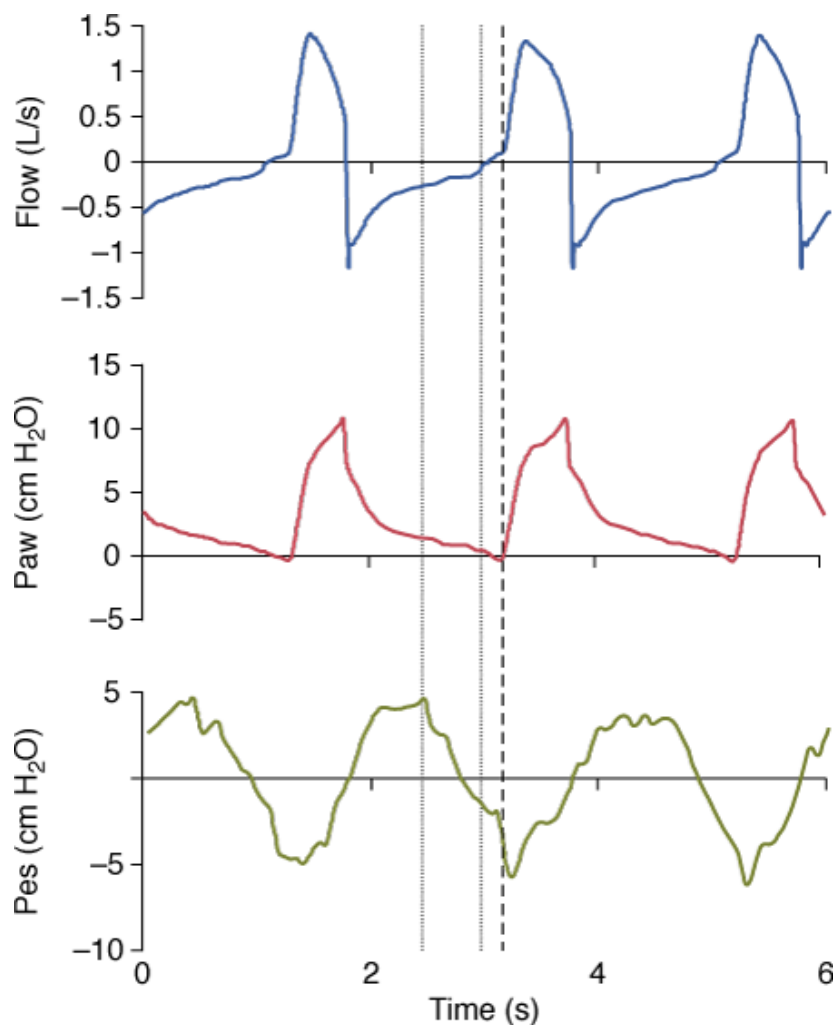


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Early cycling-off. The ventilator ends insufflation (*thick vertical dotted line*) before the patient's inspiratory effort ceases (*second vertical line*). The airway pressure tracing then drops transiently below the baseline end-expiratory pressure level because patient effort is still substantial after ventilator insufflation has ceased. The two cycles with this form of asynchrony are associated with increased effort (*dotted horizontal line*) and prolonged T_I . The duration of T_I on the second cycle (T_2) is longer than T_I on the first cycle (T_1) as reflected with the greater distance between the *arrowheads*. A large T_I favors this form of asynchrony.

Figure 8-12

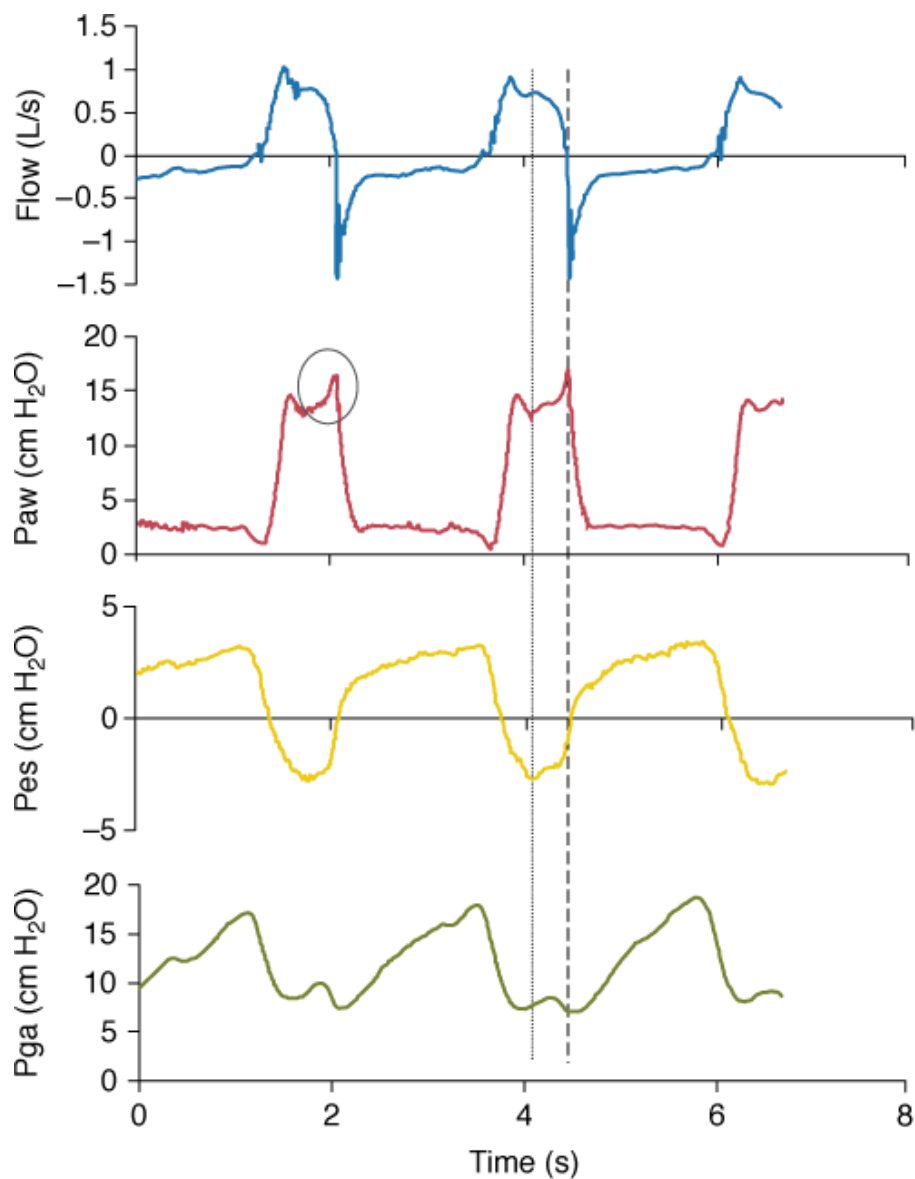


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Inspiratory and expiratory delay. Tracings in an intubated patient with severe COPD receiving PSV. An inspiratory delay secondary to auto-PEEP and triggering delay is evident. The distance between the first and second vertical dotted lines reflects the delay between onset of patient inspiratory effort and positive flow from the ventilator; this delay is caused by auto-PEEP. The delay between the second and third vertical lines is caused by triggering delay (flow triggering). In this patient, mechanical insufflation occurs almost entirely after the patient has terminated inspiratory effort. Consequently, onset of the ventilator's expiratory phase is markedly delayed compared with the patient's neural expiration.

Figure 8-13

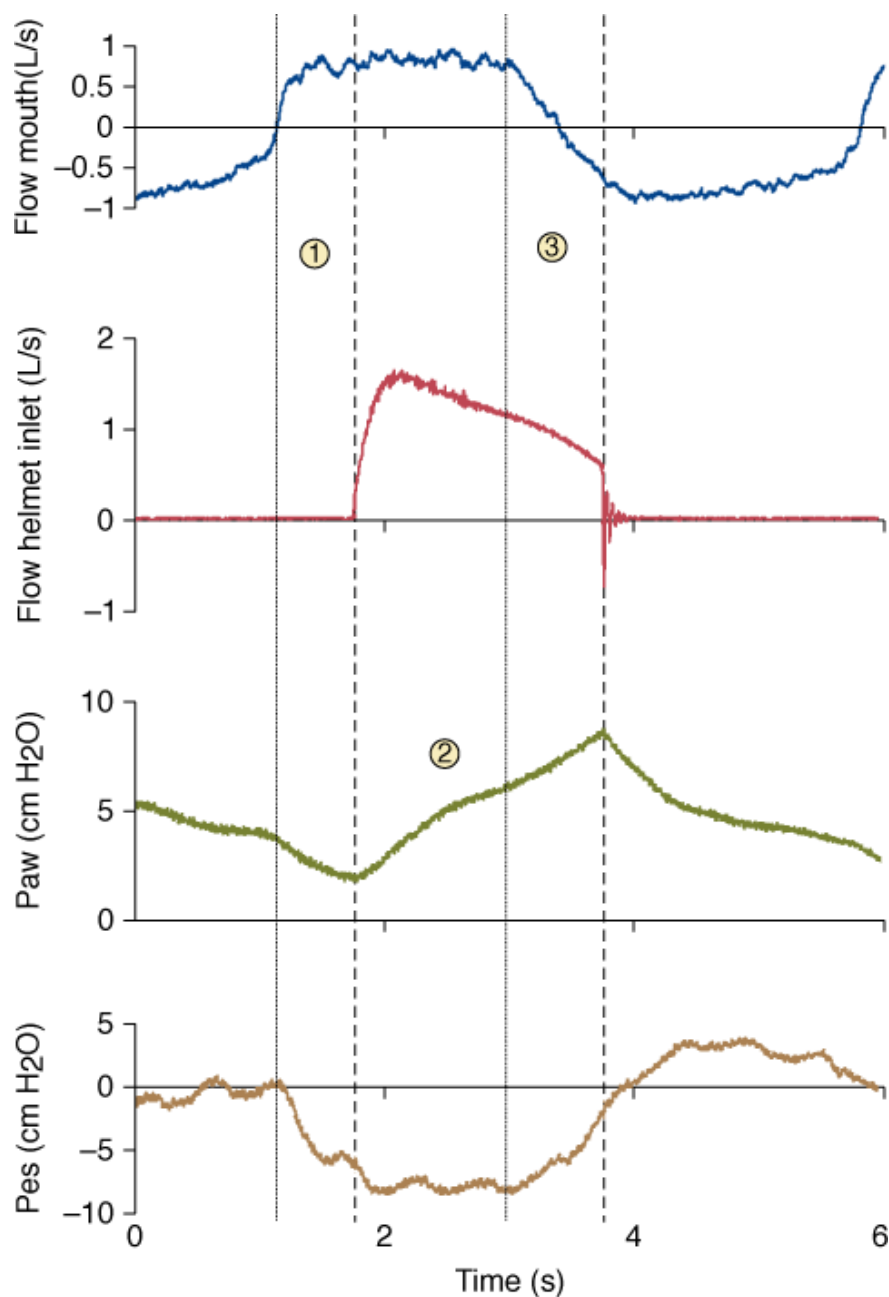


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Expiratory muscle activation. An expiratory increase in gastric pressure (*bottom tracing*) is caused by expiratory muscle activation (at the end of patient inspiration). The apparent “overshoot” on the inspiratory airway pressure tracing (*circle*) indicates, in reality, the abrupt end of patient inspiratory effort. Active expiration is present throughout all of expiration.

Figure 8-14



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Inspiratory and expiratory delays and low pressurization during PSV with a helmet system. In this healthy subject, an inspiratory delay is evident between the first vertical dotted line (onset of inspiratory effort) and the second (thicker) vertical line (onset of ventilator assistance) (interval ①). A low rate of pressurization is evident, typical when PSV is delivered via a helmet system (interval ②). Expiratory delay corresponds to the time between the third (end of inspiratory effort) and fourth vertical line (end of insufflation) (interval ③).

Synchrony between the patient and ventilator can be defined as the adequacy of matching with patient neural inspiratory and expiratory time. PSV has often been viewed as offering good synchrony because it is designed to recognize the beginning and end of each spontaneous effort. As discussed above, this is far from true in all cases. The incidence of asynchrony has been studied in three prospective studies, one in ventilator-dependent patients and two in ICU patients.^{105,109,114} These studies suggest a strong association between a high incidence of asynchronies (i.e., more than 10% of the breaths, as quantified by an asynchrony index) and a prolonged duration of mechanical ventilation.^{105,114}

During PSV, several forms of asynchrony can be identified by inspecting the airway pressure and flow curves on ventilators and it is often possible to rule out the problem by modifying ventilator settings.¹¹⁵ Many asynchronies are not specific to PSV.^{28,105} Descriptions of these asynchronies may help clinicians understand their mechanisms and thus undertake remedial steps.¹¹⁶ A parallel can be made with cardiac arrhythmias, where each type of arrhythmia has a specific treatment. We describe here the most frequent, gross, and easily recognized asynchronies encountered during PSV, according to recent descriptions.^{105,116} More subtle forms of asynchrony, such as simple delays, are difficult to detect at the bedside, and need a careful examination of esophageal pressure or diaphragm electromyogram.

At Initiation of the Cycle

Triggering delay, ineffective triggering and autotriggering are all related to lack of synchrony between onset of patient effort and onset of inspiratory assistance but results from different mechanisms (see [Figs. 8-6](#) and [8-7](#)). Short and multiple cycles (see [Figs. 8-7, 8-8, 8-9](#)) may be related to problems with inspiratory triggering, setting of the pressurization rate, and cycling criteria.

Inspiratory Trigger Delay

Trigger delay is almost inevitable with the classical triggering systems and is influenced by the system (pressure versus flow triggering) as discussed previously in the section [Differences Among Mechanical Ventilators](#). Work (or effort) in triggering the ventilator has been evaluated by comparing pressure-triggering and flow-triggering systems.^{23,27,117–120} Major delays have also been observed when using the helmet system for NIV.¹²¹ One way to decrease it is to use higher pressure levels with a faster pressurization rate.¹²²

Ineffective Triggering

This asynchrony is illustrated in [Fig. 8-6](#). During invasive ventilation with PSV, asynchrony most often results from ineffective triggering, also called wasted efforts.^{28,107–111} The frequency of this asynchrony is directly influenced by the level of PSV and dynamic hyperinflation.^{28,54,108,112,123,124} When a patient starts an inspiratory effort, a pressure gradient between the alveoli and mouth necessitates that the respiratory muscles first counteract this gradient before any inspiratory flow can be generated.²⁷ This constitutes an inspiratory threshold load, which increases breathing effort. The magnitude of positive pressure generated depends on V_T and therefore on set PSV. PSV can worsen hyperinflation by (a) delivering excessive V_T , (b) prolonging insufflation time into patient neural expiration, and (c) reducing the respiratory drive. When patient effort is feeble, effort does not reverse expiratory flow or decrease pressure sufficiently to trigger the ventilator. This produces a missed cycle. This type of asynchrony has been described mainly in patients with expiratory flow limitation and intrinsic PEEP.^{28,109,111}

In a study where assistance was varied between 0% and 100%, Leung et al found that there was almost no ineffective efforts below 60% of assistance, but they increased gradually when assistance was 60% to 100%.²⁸ In a cohort of sixty-two intubated patients, Thille et al recently found that ineffective triggering represented almost 90% of all asynchronies during PSV; a quarter of the patients exhibited a high level of asynchrony, that is, more than 10% of their efforts were asynchronous (asynchrony index greater than 10%).¹⁰⁵ COPD was a risk factor for asynchrony, as were a higher V_T , a higher setting of PSV, and progressive alkalosis.¹⁰⁵ In the study by Leung et al, cycles preceding wasted efforts were

characterized by higher V_T and lower expiratory time, which lead to greater levels of hyperinflation.²⁸ Beck et al found that the greater the level of PSV, the longer was the prolongation of insufflation into patient's neural expiration.⁵⁶ Therefore, one major reason for ineffective efforts is excessive assistance (PSV), which simultaneously generates dynamic hyperinflation and depresses respiratory drive, both because of high V_T and prolongation of insufflation far beyond the end of patient inspiratory effort.¹⁰⁵

Ineffective triggering can be detected from irregularities on airway and flow tracings during the expiratory phase (see Fig. 8-6).²⁷ A respiratory rate lower than 20 breaths/min should also rouse suspicion. Giannouli et al found that ineffective triggering could be detected as accurately on flow and airway tracings as on esophageal pressure tracings.⁵⁴ Different approaches are necessary to avoid wasted efforts: check trigger sensitivity, increase PEEP,^{108,109,123} lower PSV,^{27,28} or decrease instrumental dead space.^{92,93} External PEEP decreases the frequency of wasted efforts in some but not all studies.^{108,109,115} The two most effective approaches to decrease ineffective efforts are to decrease inspiratory time, achieved by increasing the flow cycling-off criterion or decreasing the level of PSV until the ineffective efforts disappear or the onset of signs of respiratory distress. In patients with COPD, Tassaux et al showed that it was necessary to increase the flow cycling-off criterion to 50% or 75% of the peak flow to reduce ineffective efforts.¹²⁵ Thille et al reduced the PSV level in the manner described above, such that V_T was around 6 mL/kg. This was associated with a disappearance of ineffective efforts in most, but not all, patients, without significant increase in work of breathing.¹¹⁵

Ineffective efforts exist with PSV and with ACV.¹⁰⁵ Thille et al¹⁰⁵ found that a high incidence was significantly associated with prolonged ventilation. The same association was also found by de Wit et al, and was shown to be an independent association by means of multivariate analysis.¹¹⁴ Because these asynchronies can be avoided by optimized ventilator settings, their treatment and/or prevention may result in a shorter time spent on the ventilator.

Autotriggering

Autotriggered cycles (see Fig. 8-7) are falsely triggered by a signal not coming from a patient's inspiratory effort. They can be caused by expiratory leaks around a mask during NIV, or by leaks in the ventilator circuit.¹⁰⁵ It is a special concern when using an ICU ventilator for delivering NIV.⁴² An expiratory leak can be misinterpreted by the ventilator as patient effort; an inspiratory cycle is then delivered independently of patient control. Autocycling is also caused by cardiac oscillations,¹²⁶ and when the setting trigger is excessively sensitive.

Autotriggering can be difficult to detect on the ventilator tracing. A sudden increase or a persistently high respiratory rate suggest autotriggering. The absence of airway pressure drop at the beginning of an inspiratory cycle is also suggestive (see Fig. 8-7).

Sensitivity of the inspiratory trigger can be reduced, both as a diagnostic test and as a remedy. Triggers based on flow are more sensitive than conventional triggers, and tend to increase of autotriggering.¹²⁷ A compromise must be found between triggering that is too sensitive (posing a risk of autotriggering) and too insensitive (with risk of ineffective efforts or increased effort).

Multiple Cycles

This asynchrony is illustrated in Figs. 8-8 and 8-9. Two or more ventilator insufflations may be delivered within a single patient effort. Auto-triggering can be responsible for multiple cycles (see Fig. 8-9). Ventilator characteristics, such as duration of the refractory period, may also influence this kind of asynchrony. A risk for double-triggering exists with a high inspiratory pressure ramp profile, secondary to a reduction in ventilator T_I relative to neural T_I .^{128,129} Tokioka et al described double cycles during PSV in intubated patients with restrictive lung diseases when the cycling-off criteria were high (35% and 45% of maximal inspiratory flow).¹³⁰

Three mechanisms (autotriggering, high pressurization rate, and early cycling-off) should be considered when this asynchrony is detected during PSV.

Pressurization Rate and Inspiratory Flow

The speed of pressurization determines the initial pressure ramp profile and is primarily dependent on the initial peak flow rate. This rate is adjustable on several ventilators. Altering this parameter can directly influence breathing pattern and work of breathing.¹³¹

Poor matching between patient demand and the provided peak flow may occur as a result of an inadequate rise time during PSV.^{32,59,129,132,133} Selecting a low speed of pressurization can cause excessive patient effort, especially when respiratory drive is high and mechanics are poor. Conversely, a very fast rise time may not be optimal¹³⁴ and is poorly tolerated by patients.¹³⁵ A high speed of pressurization makes it more difficult for the ventilator to properly regulate the pressure throughout inspiration according to its servo-control mechanism.

Patients with the lowest compliance and highest respiratory drive theoretically needed the highest initial flows.¹³¹ In two studies, the longer the time taken to reach the pressure level set on the ventilator, the greater the work of breathing, in patients with obstructive or restrictive lung disease.^{59,129} Excessively high pressurization can also lead to an initial overshoot, with possible early termination of the cycle related to high pressure. Chiumello et al found that the relationship between the pressurization rate and dyspnea or work of breathing exhibited a U-shape pattern.¹³⁴ During NIV, the highest pressurization rate may increase the amount of leaks and induce double triggering.¹³⁵ It can also increase respiratory rate, as previously described with ACV.^{136,137}

Inspiratory Cycling-Off or Cycling to Expiration

Delayed Occurrence of Expiration

As with delayed triggering of inspiration, this form of asynchrony is very common and related to the difference between the criterion for inspiration termination on the ventilator and the end of patient neural T_I (see Figs. 8-10, 8-12, and 8-14). Parthasarathy et al¹³⁸ studied healthy subjects with simulated airflow obstruction, and found frequent early activation of the expiratory muscles during ventilator insufflation. In intubated patients, Spahija et al compared PSV with neurally adjusted ventilatory assist (NAVA), where ventilator support is driven by the diaphragmatic electromyographic signal. Marked expiratory delays were found with high levels of PSV (1055 ± 1010 milliseconds).¹³⁹ These impressive delays may be specific to the ventilator used, Servo 300 (Maquet, Lund, Sweden), which has a low and nonadjustable cycling-off criterion (5% of peak flow).

The inspiratory termination criterion is usually a fixed percentage of peak inspiratory flow rate (12%, 25%, or 30%) and frequently does not correspond to the real end of inspiratory effort.¹⁴⁰ The effect of this setting differs between patients with obstructive and restrictive lung diseases. In general, increasing the flow criterion (expressed as a percentage of peak flow) reduces T_i .¹³¹

In patients recovering from acute lung injury, Chiumello et al found a low cycling-off criterion (5% of peak flow) was beneficial in terms of breathing pattern (reduction of respiratory rate, increase in V_T), compared with a threshold at 40%.¹³³ In similar patients Tokioka et al found an increase in V_T and decrease in respiratory rate when the cycling-off criterion was decreased from 45% to 1%.¹³⁰ These observations argue for use of a low termination criterion in patients with acute lung injury or restrictive lung disease. Conversely, in patients with COPD, the best cycling-off criterion may be above 50%.^{125,141} When the cycling-off criterion is set low (5% with the Servo 300 [Maquet, Lund, Sweden] or 5 L/min with the Puritan Bennett 7200 [Tyco, Carlsbad, CA, USA]), one can expect that insufflation will continue beyond patient neural T_i . Patients may activate their expiratory muscles, and or suffer from increased dynamic hyperinflation.^{74,138} The risk of wasted efforts during subsequent respiratory cycles increases.¹⁴²

Some patients exhibit expiratory muscle activity during PSV (see Fig. 8-13).^{74,138,143} It has been suggested that an abrupt increase in airway pressure indicates an active expiratory effort from the patient. Although this is indeed possible, a more frequent explanation is the abrupt cessation of patient active inspiratory effort. The release of this effort makes that the airway pressure suddenly reaches the preset level until the end of inspiration.¹⁴⁴

Very Prolonged Inspiration during Noninvasive Ventilation

This is illustrated in Fig. 8-10. A specific form of the preceding asynchrony may occur in case of large end-inspiratory leaks. Mechanical inflation may be prolonged far beyond the end of patient inspiration, until a limit of maximum T_i has been reached, which is sometimes adjustable. This type of asynchrony can occur during invasive ventilation but is mostly specific to leaks during NIV with ICU ventilators.¹¹³ It occurs because of the impossibility for reaching the cycling-off criterion (e.g., 25% or less of peak inspiratory flow). In the case of end-inspiratory leaks around the mask or in the circuit, the ventilator continues insufflation and flow does not decrease (because of these leaks). The breath does not terminate and is prolonged until a maximum T_i is reached (which can be several seconds). The patient may “fight” against the ventilator and may even attempt additional inspirations. Ineffective efforts can be observed in this context, either within the same ventilator cycle or in following cycles.

Prolonged inspiration is a common asynchrony in patients receiving NIV for acute respiratory failure with ICU ventilators not having a noninvasive mode.¹⁴⁵ The first step is to reduce leaks and inspiratory pressure by decreasing PEEP or PSV. This asynchrony can also be avoided by adjusting the cycling-off criterion higher. Calderini et al set T_i between 0.8 and 1.2 seconds.¹¹³ This produced better matching between the patient and the ventilator, reducing work of breathing and improving patient comfort.¹¹³ Assist-control pressure ventilation can also be used to deliver pressure with a fixed T_i . Large delays may also result with use of a helmet for NIV (see Fig. 8-14).¹⁴⁶ New ICU ventilators using the NIV mode and dedicated NIV ventilators generally prevent, to a certain extent, the occurrence of these prolonged insufflations.^{40,145}

Early Cycling-off

This asynchrony is illustrated in Figs. 8-7, 8-8 and 8-11. Very short, aborted cycles can occur during NIV and invasive ventilation. This may indicate either a switch-off mechanism that occurs too early or autotriggering (see Fig. 8-7).¹⁰⁵

If the cycling-off criterion is reached too early, the ventilator stops insufflation and opens the expiratory valve while patient inspiratory effort continues. This produces an initial drop in airway pressure and flow, followed by an increase related to patient inspiration, resulting in a characteristic contour (see Fig. 8-11).¹³⁰ This asynchrony is observed during invasive ventilation and NIV. In patients recovering from acute lung injury,¹³³ the cycling-off criterion may be set lower to minimize this risk.

Yamada and Du¹⁴¹ designed a mathematical model to analyze the mechanisms of expiratory asynchrony during PSV.¹⁴¹ The ratio of flow at the end of patient neural inspiration (neural T_I) to peak inspiratory flow during PSV is determined by the ratio of respiratory time constant (τ) to neural T_I and by the ratio of set PSV to maximal inspiratory muscle pressure. They found that with selected respiratory mechanics, the ratio of flow at the end of neural T_I to peak inspiratory flow ranged from 1 to 85%, and had an excellent linear correlation with the τ -to-neural T_I ratio. The highest values of the cycling-off criterion corresponded to obstructive patients with high resistances and high compliances (resulting in a high time constant). The lowest values corresponded to patients with acute lung injury, who have low resistances and low compliance (resulting in a short time constant).

Hotchkiss et al used linear and nonlinear mathematical models to investigate the dynamic behavior of PSV. Predicted behavior was confirmed with a test lung.¹⁴⁷ In the setting of airflow obstruction, PSV was accompanied by marked variations in V_T and end-expiratory alveolar pressure, even when patient effort was unvarying. Unstable behavior was observed in the simplest plausible linear mathematical model, and it was an inherent consequence of the underlying dynamics of this mode. Because of its complexity and the frequent changes in ventilatory pattern during PSV,¹⁴⁷ automatic adjustment of triggering, based on mathematical models, might be helpful.¹⁴⁸ Du et al proposed an automatic adjustment of the cycling-off criterion based on the measured time constant in a patient.^{149,150}

Clinical Detection

Major asynchronies can be detected at the bedside on the ventilator screen by looking at airway pressure and flow tracings.¹¹⁶ New ventilators permit clinicians to adjust ventilator settings in order to reduce the frequency of asynchrony.¹¹⁶ During invasive PSV, a major goal is recognition of ineffective efforts. During noninvasive PSV, an important issue is recognition of prolonged inspirations, caused by leaks, or autocycling.⁴² Adjustment of settings (simple decrease of pressure level or change in the cycling-off criterion) avoids or minimizes the problem. This step is likely to improve the efficacy and comfort of PSV.

Differences from Other Modes of Ventilation

Intermittent Positive-Pressure Breathing

PSV has some similarities with intermittent positive-pressure breathing (IPPB), an assisted mode used widely in the 1960s for physiotherapy.^{151–154} With both IPPB and PSV, cycles are triggered by the patient and limited by pressure, and both can assist patients in acute respiratory failure (with or without endotracheal intubation). They differ in that

PSV, but not IPPB, maintains a constant level of pressure during inspiration. The mechanisms that cycle between inspiration and expiration are also different. The end of inspiration is flow-cycled with PSV and is pressure-cycled with IPPB. In normal, nonintubated patients, work of breathing was lower with PSV than with IPPB,¹⁵⁵ a difference considerably exaggerated in the presence of CO₂ stimulation. Expiratory work was also greater and comfort poorer with IPPB devices than with PSV.

Assist-Control Ventilation

Both PSV and volume-controlled ACV can provide full ventilator support but ACV is used more frequently, especially during the early phase of mechanical ventilation because it offers a more stable ventilation with regards to changes in respiratory mechanics.⁷ The modes differ in ways that explain their relative advantages and disadvantages. During ACV, V_T is guaranteed and independent of respiratory mechanics, and a minimal frequency and minute ventilation is set. During PSV, by contrast, V_T may change with alterations in respiratory system compliance or resistance.

For any inspiratory effort, addition of PSV augments the pressure difference between the circuit and the alveoli, leading to a higher inspiratory flow rate and higher V_T than during spontaneous breathing. Until high PSV levels are reached, it increases inspiratory flow rate in a way that remains partially under patient regulation. During ACV, patient inspiratory effort does not modify flow or volume. Therefore, the use of volume ACV is essential when strict control of V_T or transpulmonary pressure is considered important, such as acute respiratory distress syndrome to avoid excessive distension.^{156,157} Use of PSV in unstable patients with a high respiratory drive has the major disadvantage of offering no control over V_T , and the pressure level used does not give any indication of what is the real transpulmonary pressure.¹⁵⁸ Conversely, PSV may adapt better to variation in patient demand.

Effects of PSV and ACV on breathing pattern, gas exchange, and indexes of work or effort have rarely been compared. Cinnella et al compared breathing pattern and respiratory muscle effort during ACV and assisted pressure-control ventilation.¹⁵⁹ Although the latter mode differs from PSV in that T_I is preset, this comparison enabled the study of the comparison of a pressure-targeted and a flow-targeted mode. Pressure was adjusted to achieve a similar V_T and T_I as with ACV. With a high V_T (12 mL/kg), the modes did not differ for respiratory muscle effort. At a moderate V_T (8 mL/kg), the decelerating flow pattern of the pressure-targeted mode better matched patient demand than did the constant-flow pattern and work of breathing and pressure-time index were significantly lower with assisted pressure control ventilation. This difference was abolished, however, when inspiratory peak flow rates were increased. Thus, with adequate settings, both modes could adequately unload the muscles. Leung et al also showed that high levels of assistance was equivalent with PSV, ACV, and SIMV.²⁸

Synchronized Intermittent Mandatory Ventilation

SIMV combines delivery of assisted breaths with spontaneous unassisted breaths. SIMV differs from PSV, where every breath is supported to the same extent. On a breath-to-breath basis, the effort performed by a patient during SIMV is almost equivalent for assisted and unassisted breathing,^{160,161} although differences between assisted and unassisted breaths have been found during SIMV when the mandatory breaths were delivered as pressure-targeted breaths.²⁷

MacIntyre compared SIMV (V_T set at 10 to 15 mL/kg), and PSV (set at 13 to 41 cm H₂O) in a crossover study of fifteen patients recovering from acute respiratory failure.¹¹ Sense of comfort was increased and respiratory rate slower with

PSV. In patients ventilated for at least 3 days, Knebel et al compared similar levels of partial support provided by SIMV and PSV in terms of breathing comfort, defined by subjective ratings of dyspnea and anxiety.¹⁶² Preweaning levels of dyspnea and anxiety did not differ significantly between the modes at any level of support. Comfort was not influenced by the level of support, and was similar with the two modes.

Leung et al compared the effects of PSV, SIMV, and their combination at varying levels of support in the same patients.²⁸ Patient effort was similar with SIMV and PSV at high levels of assistance, but was higher with SIMV than with PSV at lower levels of assistance (20% to 40% of maximal support). The same difference between the two modes was found in animal experiments.¹⁶³ The studies suggest that when reducing the level of ventilator support, unloading of the respiratory muscles occurs earlier with SIMV than with PSV.¹⁶⁴ Whether this could explain the different clinical outcomes with the two modes during weaning is unclear.

Proportional-Assist Ventilation

PAV was developed several years after PSV.^{165,166} Promising initial results and great physiologic interest has characterized this mode. PAV, however, has essentially remained a physiologic tool, and results of the first clinical comparisons with PSV have been disappointing. Most of physiologic comparisons of PAV and PSV favor PAV in terms of breathing pattern variability, patient-ventilator interaction, and comfort. In clinical trials, mainly during NIV, these physiologic effects did not produce any outcome benefit until recent developments.¹⁶⁷⁻¹⁷¹

PAV is designed to deliver assistance in direct proportion to patient effort.^{165,166,172}

Advantages of Proportional-Assist Ventilation Compared with Pressure-Support Ventilation

During invasive ventilation and NIV, PAV is consistently superior to PSV on physiologic end points. In response to variable loads (dead space, resistive or restrictive loads), PAV adapts the level of assistance to patient demand. With PSV, in contrast, the level of inspiratory pressure remains constant whatever the load. In response to acute hypercapnia, Ranieri et al showed that PAV adapted more efficiently to ventilatory demand than did PSV.¹⁷³ Levels of inspiratory pressure increased during PAV with a relative increase in V_T and no change in respiratory rate, while a small increase in V_T and a large increase in rate were observed with PSV. This resulted in lower work of breathing and better comfort with PAV.¹⁷³ In intubated patients, the response to a restrictive load (chest and abdominal binding) resulted in less work and better comfort when compared with PSV.¹⁷⁴ Such adaptations to loads with PAV and PSV were also found in healthy subjects.^{175,176}

Several studies found less ineffective triggering with PAV than with PSV in intubated patients.^{54,177-179} With PAV, ventilator T_I exactly matches neural T_I , even in patients with a long time constant. Thus, insufflation continues while the patient has begun to exhale with PSV, but not with PAV. Many studies also reveal a greater variability of V_T with PAV than with PSV,^{176,180-183} and better patient comfort.^{167,168,173,176,181,184-188}

Clinical Trials

Two prospective randomized controlled studies with clinical end points compared PAV and PSV during NIV.^{167,168} The first study enrolled forty-four patients and the second enrolled 117 patients.¹⁶⁸ Intubation and mortality rates were

equivalent despite better comfort and less intolerance with PAV. These early clinical trials of PAV are disappointing when contrasted with the promising physiologic studies.

Proportional-Assist Ventilation with Load-Adjustable Gain Factor

A new method now enables automatic measurement of respiratory mechanics with PAV. The initial results are encouraging.^{189,190} In contrast with any other mode (including PSV), greater knowledge of resistive and elastic characteristics of the respiratory system is necessary when setting PAV. It is often difficult to obtain simple and reliable measurements in awake patients triggering the ventilator and the values frequently vary. The setting of PAV is thus complex, and constitutes an obstacle to its wide acceptance. The time needed to set PAV is much longer than for PSV.¹⁸² Also, a major drawback of PAV is the occurrence of a specific asynchrony: flow and pressure runaway.¹⁷² Runaway is mainly related to an excess of volume and elastic assistance because elastance is overestimated.^{54,177,184} Methods to automatically and intermittently determine respiratory resistance and elastance were recently described.^{189,190} These methods could be incorporated into a closed-loop adjustment of PAV, and possibly constitute a major step forward. A potentially important advance regarding the clinical use of PAV is the possibility to get automated reliable repeated measurements of elastance and resistance of the respiratory system. Few clinical comparisons exist, again mostly with PSV. The effect of modifications in the respiratory loading conditions were compared between PAV with load-adjustable gain factor, referred to as PAV+, against PSV.¹⁹¹ Adaptation was more physiologic with PAV+, allowing patients to keep the same $\dot{V}_T$ and minute ventilation, by contrast with PSV. One clinical evaluation of PAV+ randomly compared it to PSV for 48 hours of assistance.¹⁹² In this study, failure to maintain assisted spontaneous ventilation was higher with PSV than with PAV+, and PAV+ was associated with a major reduction of ineffective efforts.

Neurally Adjusted Ventilatory Assist

NAVA is a promising mode of ventilation, although it still lacks clinical data to precisely define its clinical application. Like PAV, it provides assistance in proportion to patient effort but it depends on continuous recording of Edi, which is obtained via a nasogastric catheter incorporating a multiple array esophageal electrode (nine electrodes spaced 10 mm apart). The ventilator thus acts like a muscle under a patient's neural command. Inspiratory airway pressure applied by the ventilator is determined by the following equation:

$$Paw(t) = Edi(t) \times \text{NAVA level}$$

where $Paw(t)$ is the instantaneous airway pressure (cm H₂O), $Edi(t)$ is the instantaneous diaphragmatic electrical activity signal (μV), and NAVA level (cm H₂O/μV or per arbitrary unit) is a proportionality constant set by the clinician.

The onset and end of assistance and the level of assistance are directly driven by the Edi signal.¹⁹³ In theory, NAVA should provide much better patient-ventilator synchrony than achieved with pressure-targeted modes, and both experimental and clinical results support this expectation. Unlike all other modes (including PAV), NAVA should not be influenced by intrinsic PEEP or by the presence of leaks as in the case of standard triggering systems. The initial reports on NAVA revealed advantages compared with PSV in terms of triggering and cycling-off of the ventilator.¹⁹³ In a rabbit model of acute lung injury,¹⁹⁴ diaphragmatic unloading was much more efficient with NAVA than with PSV, with an absence of wasted efforts. In intubated patients, Spahija et al compared the trigger delay and cycling-off of inspiration with NAVA and PSV¹³⁹ at low and high levels of assistance. Inspiratory trigger delay was around 100 milliseconds with

NAVA and around 200 milliseconds with PSV. Cycling-off delays were markedly different: 40 milliseconds (whatever the level of assistance) with NAVA and 500 to 1000 milliseconds with PSV (low and high assistance). The same group also reported an evaluation of an original closed-loop system, using Edi as a target to select the level of PSV: target-drive ventilation.¹⁹⁵ This system was evaluated in eleven healthy subjects, before and during exercise, and without ventilator support. Without target-drive ventilation, Edi increased, as did indexes of effort and end-tidal CO₂ during exercise. With target-drive ventilation, the level of pressure increased during exercise, maintaining the Edi constant.

In the more recent comparisons with PSV, NAVA reproducibly decreased triggering delay by more than 50%, in addition to significantly reducing cycling delay and total asynchrony events.^{139,196} Terzi et al showed that beyond a pure proportional assistance, Edi triggering further decreases patient-ventilator asynchrony.¹⁹⁶ These studies also demonstrated that NAVA limits overassistance in contrast to PSV. NAVA prevents overassistance because excessive assist downregulates respiratory center activity (hence Edi) and, thus, inspiratory pressure. NAVA reveals the patient's natural breathing variability that is otherwise masked by constant PSV assist.¹⁹⁷ This finding could be relevant in explaining the oxygenation improvement with NAVA compared to PSV in a 24-hour crossover study of twelve postoperative patients.¹⁹⁸ Thus, the most recent studies have confirmed that NAVA improves patient-ventilator interaction compared to PSV and prevents overassistance. Nevertheless, it remains to be demonstrated that providing control over breathing pattern to a patient will improve clinical outcome.

Hemodynamic Consequences of Pressure-Support Ventilation

Unique hemodynamic consequences have not been described with PSV.^{11,12,45,47,63,76,199} Despite a wide range of pressures, most studies have found little or no deleterious effects of PSV on cardiovascular function in patients after cardiac surgery or in patients with respiratory failure. The negativity of pleural pressure during PSV and the control of airway pressure reduces the risk of negatively influencing venous return or right ventricular afterload. One study found that PSV during NIV may lead to a reduction in cardiac output without change in mixed venous.⁶⁵ Whether this reflects adaptation to a decrease in CO₂ production or a deleterious effect could not be inferred from these data.²⁰⁰

In patients with cardiac dysfunction, it was shown that weaning-induced left-ventricular dysfunction appears sooner during T-piece trials than during PSV trials because the workload is higher during the T piece.⁸⁹ Whether T piece better predicts postextubation problems in cardiac patients cannot be directly inferred from these data.

Adjustment of Pressure Level at Bedside

Precise guidelines for the bedside use of PSV are lacking because the pressure level has been adjusted in various ways in many studies.^{11,201,202} Recent data reinforce the notion that targeting a V_T lower than traditionally used and a higher respiratory frequency offers benefit for the patient.^{69,115} Assessment of accessory muscle activity, especially the sternocleidomastoid, by inspection and palpation was suggested for deciding optimal assistance.⁴³ A decrease in respiratory frequency below 30 to 32 breaths/min was also associated with an optimal level of PSV. Respiratory frequency can be used as a simple indicator of the adequacy of PSV^{43,76}; and less than 30 breaths/min has been recommended.^{74,76,203} As indirectly suggested by weaning trials, targeting a low frequency (25 breaths/min or less)

can prolong weaning duration. The latter may also be explained by increased occurrence of asynchrony when a low frequency is displayed on the ventilator.

In early studies, PSV was adjusted to reach a predetermined V_T (8 to 12 mL/kg).^{12,47,70} A setting of closer to 6 mL/kg V_T seems more advisable to avoid patient-ventilator dyssynchrony. One study suggested, however, that patients who could be weaned controlled their own V_T and were only mildly influenced by the PSV level.²⁰⁴

In patients who display major asynchronies, such as ineffective efforts, reducing the level of pressure support¹¹⁵ and reducing the insufflation time through an increase in the flow-cycling criterion¹²⁵ are recommended. This leads to a smaller V_T than usually recommended, around 6 mL/kg. Interestingly, a recent study using the DLCO technique showed that gas exchange was overall better with a V_T of 6 mL/kg versus 8 mL/kg.⁶⁹

Closed-Loop Delivery of Pressure-Support Ventilation

Dual Modes

The potential for variation in delivered ventilation has led many manufacturers to develop servo-controlled modalities of PSV. In one ventilator, the PSV level was automatically adjusted to achieve a preset breathing frequency.²⁰⁵ Several servo-controlled modes have been proposed to adjust the PSV level to keep V_T constant. Pressure can be varied from breath to breath using various algorithms. These options have been designed to provide a better ventilator response to changes in respiratory mechanics.^{206,207} An increase in resistance or elastance during PSV normally leads to a drop in V_T if no compensation is made by the patient or the ventilator.

These modes, often called *dual-control modalities*, use closed-loop feedback control systems that enable the ventilator to adapt output based on the difference between measured ventilation and a predefined target. The modes go by different names. Volume-support ventilation (VSV) was introduced in the 1990s on the Servo 300 ventilator (Siemens Elema, Solna, Sweden). VSV is a pressure-limited mode that uses a target V_T and minute ventilation for feedback control.

The level of PSV is adjusted continuously to deliver a preset V_T . Two anecdotal reports with VSV^{208,209} and one randomized controlled trial²¹⁰ have yielded variable results. Several modes are now working on the same principle—volume targeted pressure-regulated mode—and have been extensively developed by manufacturers without clinical validation. Physiologic studies have not assessed the efficacy of VSV in terms of adjustment to spontaneous changes in mechanics. The response of such modes to changes in ventilatory demand can be problematic. Changes in demand frequently occur with different states of, for example, wakefulness, nutrition, episodes of sepsis, pain, and anemia. With a fixed level of VSV, but not of PSV, it was shown that an increase in ventilatory demand resulted in a decrease in the level of support provided by the ventilator, the opposite of a desired response.^{211–213} Conceivably, VSV may result in respiratory distress in clinical settings.²¹¹ In general, these modes are not able to cope correctly with changes in ventilator demand. They also make rapid alterations in the level of support, thus potentially interfering with a patient's own response time.⁵⁰

Knowledge-Based Systems

More complex knowledge-based systems have been developed with the aim of providing an automatically performed, patient-adapted ventilator support, which is superimposed on an automated weaning strategy.^{104,214–216} Such systems have been implemented in computers that drive a ventilator. These approaches using SIMV plus PSV or PSV alone have been evaluated in patients. Sophisticated modes have been developed through improvements in computer science.^{217,218}

When the physiologic and clinical knowledge needed to manage a well-defined clinical situation is acquired, it can be embedded within a computer program that drives the ventilator using artificial intelligence techniques, such as production rules, fuzzy logic, or neural networks.²¹⁹ These techniques allow planning and control. Control is a local task, which consists of determining what the immediate next step is. Planning is a strategic task, aimed at regulating the time-course of the process. For control and planning, numerous techniques have been developed in the fields of control theory and artificial intelligence, respectively. The main difference between these two fields lies in the process models used. It is important to avoid both oversimplification and excessive complexity.²⁰⁹ Strickland and Hasson tried to develop a controller incorporating an active clinical strategy represented by production rules using SIMV and PSV (IF conditions, THEN actions). Their work did not lead to commercial development.^{215,216}

The Smart Care system is an embedded version of the initial NeoGanesh system. The NeoGanesh system drives the ventilator with PSV, keeping a patient within a zone of “respiratory comfort” as defined by respiratory parameters, and superimposing an automated strategy for weaning.^{104,214,220} The designers of the knowledge-based NeoGanesh system intended to build a closed-loop system that (a) was efficient for automatically controlling PSV and planning of the weaning process, (b) could be evaluated with the goal of gradually improving its reasoning and planning capabilities, and (c) could be subjected at the bedside to performance measurements at each step of its operation. The NeoGanesh system is based on modeling of the medical expertise required to perform mechanical ventilation in PSV mode. It does not include mathematical equations of a physiologic model. Several types of evaluation have been performed: (a) to determine how well the system adapts the level of assistance to patient needs (evaluation of the control level),²²⁰ (b) to assess the extubation recommendation made by the system (evaluation of the strategic level),²¹⁴ and (c) to estimate the impact on clinical outcomes.^{221,222} This system reduces periods of excessive respiratory efforts and predicts extubation time with good accuracy.^{214,220} It has been used safely during prolonged periods of mechanical ventilation and often predicted earlier than clinicians the time at which patients were ready to be separated from the ventilator.²²¹ A multicenter study comparing this system of automated weaning to usual weaning was later performed.²²² Five academic centers recruited 144 patients in 1 year; patients were included as soon as they could tolerate PSV. It was compared to usual care, as performed in the various ICUs participating in the study. Weaning duration was reduced by a median of 2 days, which resulted in a shorter duration of mechanical ventilation and ICU stay. These impressive results have not been consistently reproduced. A smaller randomized study showed no difference with a strict weaning approach,²²³ whereas other reports a reduction of weaning duration in surgical patients.²²⁴

Noisy Pressure-Support Ventilation

Introducing a random noise on the distribution of pressure during PSV produces interesting physiologic effects.^{225–227} In two animal models of acute lung injury, noisy PSV increased the variability of the respiratory pattern and improved oxygenation by a redistribution of perfusion toward the ventilated nondependent lung regions.^{225,226}

Predicting the Effect of Pressure-Support Ventilation Based on Load Estimation

It has been suggested that a noninvasive method allowing load estimation could be used to titrate the level of pressure support.²²⁸ Noninvasive measurement of the power of breathing, and tolerance of these loads, reflected by spontaneous breathing frequency and V_T , could be considered for deciding the level of PSV, so that muscle loads are not too high or too low. Such a computerized PSV advisory system provided recommendations for setting PSV to unload the inspiratory muscles that were essentially the same as the recommendations from experienced, critical care respiratory therapists.²²⁹

Clinical Applications

Weaning

The usual weaning methods are once-daily trials of spontaneous breathing, most often with a T piece, resulting in abrupt discontinuation of mechanical ventilation, SIMV, and PSV, with a gradual reduction in the level of assistance. A low PSV level can be used to mimic spontaneous breathing trials, as mentioned earlier ([Compensation for the Work Caused by Endotracheal Tube and Demand Valve](#)); the latter trial thus constitutes the final step in the approach of gradual reduction in PSV or as a substitute for a T-piece trial. PSV can also be used in combination with SIMV, although very few data support the use of this combination.^{26–28} In a 1998 international survey of the use of mechanical ventilation,⁷ PSV was used one way or another in 45% of weaning attempts, indicating that PSV is considered an important weaning technique.

Studies comparing T-piece trials, SIMV, and PSV were rare before the mid-1990s. Studies had included a large percentage of postoperative patients who exhibited no persistent weaning problem; no conclusions were drawn regarding the type of support to use.²³⁰ PSV as a sole mode of ventilation has first been tested in two large prospective, randomized, controlled trials.^{201,202} These trials share common conclusions: (a) patients who tolerated 2 hours of breathing on a T-piece constituted 60% to 80% of patients and were easily separated from the ventilator on first attempt; (b) weaning outcome in the remaining patients depended heavily on the weaning strategy used; and (c) SIMV was consistently the worst weaning method. The two trials, however, differed regarding efficacy of PSV. PSV was found superior to other methods in the study of Brochard et al, but not in the subsequent study of Esteban et al.

At 21 days, a significantly higher percentage of patients had been separated from the ventilator with PSV than with the other two methods in the trial of Brochard et al,²⁰¹ and this was accompanied by a shorter duration of weaning with PSV. Esteban et al adjusted the pressure level to achieve a respiratory rate lower than 25 breaths/min, much lower than the 35 breaths/min upper limit in the trial of Brochard et al.²⁰² PSV was found to be inferior to once-daily T-piece trials and multiple daily T-piece trials in the trial of Esteban et al. This PSV approach seemed to lengthen the weaning process compared with once-daily T-piece trials. In the study of Esteban et al, the final test before extubation was a T-piece trial, during which respiratory rates of up to 35 breaths/min were tolerated. In the same study, the final step with PSV was a level of 5 cm H₂O, during which respiratory rates above 25 breaths/min were considered a sign of poor clinical tolerance. Thus, the use of PSV differed markedly in the two studies and likely explains differences in efficacy. No study has compared the PSV approach in the trial of Brochard et al to the once-daily T-piece method in the study of Esteban et al. Another trial, however, found no difference between T-piece trial and PSV in patients with prolonged weaning difficulties.²³¹

PSV and T piece have been compared as a final test before extubation. Use of a low level of PSV has been found equivalent or slightly superior to a T-piece trial both in adults and infants.^{85,86,232} A low level of PSV was slightly superior to T piece in terms of short-term success after extubation.^{85,232} At 48 hours, the success rate was not significantly different in the largest trial.⁸⁵ A small study suggested that patients who do not tolerate a T-piece trial may tolerate a low level of PSV without increasing the risk of reintubation.⁸⁸ The study was too small to render definite conclusions, but concurs with recent data showing that the T-piece trial imposes a higher load on the respiratory muscles.⁸⁹

A comparable approach was used in the NeoGanesh closed-loop system that employs automatic adjustment of PSV. Although different from standard clinical use of PSV, the results of computer-driven system constitute a form of validation of PSV weaning, in a system that applies it on a 24-hour-a-day basis.^{222,233}

Noninvasive Ventilation

NIV has been used in many patients with acute respiratory failure since the early 1990s.^{8,9,36,110,146,234–240} PSV is used preferentially for NIV. Delivered via a face mask, PSV markedly decreases respiratory muscle activity in patients with COPD in acute exacerbation.^{235,241} During exercise, which also constitutes a high ventilatory workload, noninvasive PSV improves performance in patients with COPD.²⁴²

Bi-level ventilation, as PSV is often called when combined with PEEP, is by far the most frequent mode of ventilation. In a large, multicentric French survey of NIV in ICUs, PSV was used in 67%, ACV in 15%, and continuous positive airway pressure in 18% of patients.²⁴³ PSV is often considered better tolerated than ACV during NIV.²⁴⁴

Determinants of the success of NIV have been described in several studies. They relate to patient characteristics (severity scores, etiology of the respiratory failure, nutritional status)^{111,241,243} and immediate outcome variables (evolution of arterial blood gases over 2 hours, comfort, level of leaks).^{111,243,245–248} These determinants are in part related to technical aspects: ventilator performance, modes,^{110,244} settings,^{20,31,110,113,123,249} interfaces,^{146,250,251} and patient-ventilator asynchrony.^{113,145}

Use of Noninvasive Ventilation with Pressure-Support Ventilation for Weaning

In patients with acute exacerbations of COPD, prolonged mechanical ventilation is associated with complications. In this setting, Nava et al suggested that deliberate extubation followed by a switch to NIV might improve weaning outcome.²³⁹ Using PSV for such a goal achieved greater weaning success and higher survival rate. A subsequent study did not confirm these results; patients switched to noninvasive PSV experienced a longer time of ventilator support.²⁵² In patients with stable chronic respiratory disorders who were unable to sustain spontaneous breathing, Vitacca et al found that invasive PSV, delivered while patients were still intubated, and noninvasive PSV were equally effective in reducing respiratory muscle work and improving arterial blood gases.²⁵³ In addition, noninvasive PSV was slightly better tolerated.

The best indication for NIV with PSV after extubation seems to be the prevention of subsequent respiratory failure in patients at risk of reintubation, as shown in several randomized clinical trials.^{254,255} By contrast, applying NIV at the time of postextubation respiratory distress has not proved efficient and can even delay reintubation.^{256,257}

Conclusion

PSV is a mode of partial ventilator assistance that has proved very efficient in reducing work of breathing and can offer often acceptable synchrony with patient effort. As such, PSV has constituted an enormous change in the way patients have been ventilated over the last 15 years, and has also provided a model to increase understanding of patient-ventilator interaction. Its place has been first assessed during weaning and NIV but is by now largely increasing. It can be used early in the course of ventilation, enabling the patient to be ventilated without need for sedation and in preparation for weaning. In the early phase, its use was limited by lack of control over the volume delivered and the lack of knowledge of the transpulmonary pressure generated. One of its main drawbacks, however, is the possibility of overassisting patients, causing dyssynchrony between a patient's rhythm and that of the ventilator, inducing hyperventilation, episodes of apneas and potentially sleep fragmentation, especially in patients with obstructive lung disease. For clinical use, lower tidal volumes than usually recommended and relatively higher frequencies should be targeted together with clinical assessment of patient comfort.

References

1. Brochard L. Less sedation in intensive care: the pendulum swings back. *Lancet*. 2010;375:436–438. [[PubMed: 20116840](#)]

2. Strom T, Martinussen T, Toft P. A protocol of no sedation for critically ill patients receiving mechanical ventilation: a randomised trial. *Lancet*. 2010;375:475–480. [[PubMed: 20116842](#)]

3. Sassoone CS, Zhu E, Caiozzo VJ. Assist-control mechanical ventilation attenuates ventilator-induced diaphragmatic dysfunction. *Am J Respir Crit Care Med*. 2004;170:626–632. [[PubMed: 15201132](#)]

4. Levine S, Nguyen T, Taylor N, et al. Rapid disuse atrophy of diaphragm fibers in mechanically ventilated humans. *N Engl J Med*. 2008;358:1327–1335. [[PubMed: 18367735](#)]

5. Jung B, Constantin JM, Rossel N, et al. Adaptive support ventilation prevents ventilator-induced diaphragmatic dysfunction in piglet: an in vivo and in vitro study. *Anesthesiology*. 2010;112:1435–1443. [[PubMed: 20460996](#)]

6. Boles JM, Bion J, Connors A, et al. Weaning from mechanical ventilation. *Eur Respir J*. 2007;29:1033–1056. [[PubMed: 17470624](#)]

7. Esteban A, Anzueto A, Frutos F, et al. Characteristics and outcomes in adult patients receiving mechanical ventilation: a 28-day international study. *JAMA*. 2002;287:345–355. [[PubMed: 11790214](#)]

8. Esteban A, Ferguson ND, Meade MO, et al. Evolution of mechanical ventilation in response to clinical research. *Am J Respir Crit Care Med*. 2008;177:170–177. [[PubMed: 17962636](#)]

9. Esteban A, et al. Third international mechanical ventilation study. 2011. Personal communication.

10. Kacmarek RM. The role of pressure support ventilation in reducing work of breathing. *Respir Care*. 1988;33:99–120.

11. MacIntyre NR. Respiratory function during pressure support ventilation. *Chest*. 1986;89:677–683. [[PubMed: 3698697](#)]

12. Prakash O, Maij S. Cardiopulmonary response to inspiratory pressure support during spontaneous ventilation vs conventional ventilation. *Chest*. 1985;88:403–408. [[PubMed: 3875454](#)]

13. Brochard L, Pluskwa F, Lemaire F. Improved efficacy of spontaneous breathing with inspiratory pressure support. *Am Rev Respir Dis*. 1987;136:411–415. [[PubMed: 3619200](#)]

14. Katz JA, Kraemer RW, Gjerde GE. Inspiratory work and airway pressure with continuous positive airway pressure delivery systems. *Chest*. 1985;88:519–526. [[PubMed: 3899530](#)]

15. Marini JJ, Capps JS, Culver BH. The inspiratory work of breathing during assisted mechanical ventilation. *Chest*. 1985;87:612–618. [[PubMed: 3987373](#)]

16. Beydon L, Chassé M, Harf A, Lemaire F. Inspiratory work of breathing during spontaneous ventilation using demand valves and continuous flow systems. *Am Rev Respir Dis*. 1988;138:300–304. [[PubMed: 3057961](#)]

17. Samodelov LF, Falke KJ. Total inspiratory work with modern demand valve devices compared to continuous flow CPAP. *Intensive Care Med*. 1988;14:632–639. [[PubMed: 3053843](#)]

18. Cox D, Tinloi SF, Farrimond JG. Investigation of the spontaneous modes of breathing of different ventilators. *Intensive Care Med*. 1988;14:532–537. [[PubMed: 3221008](#)]

19. Richard JC, Carlucci A, Breton L, et al. Bench testing of pressure support ventilation with three different generations of ventilators. *Intensive Care Med*. 2002;28:1049–1057. [[PubMed: 12185424](#)]

20. Lofaso F, Brochard L, Hang T, et al. Home vs intensive-care pressure support devices: experimental and clinical comparison. *Am J Respir Crit Care Med*. 1996;153:1591–1599. [[PubMed: 8630607](#)]

21. Stell I, Paul G, Lee K, et al. Noninvasive ventilator triggering in chronic obstructive pulmonary disease. A test lung comparison. *Am J Respir Crit Care Med*. 2001;164:2092–2097. [[PubMed: 11739140](#)]

22. Thille AW, Lyazidi A, Richard JC, et al. A bench study of intensive-care-unit ventilators: new versus old and turbine-based versus compressed gas-based ventilators. *Intensive Care Med*. 2009;35:1368–1376. [[PubMed: 19352622](#)]

23. Aslanian P, El Atrous S, Isabey D, et al. Effects of flow triggering on breathing effort during partial ventilatory support. *Am J Respir Crit Care Med*. 1998;157:135–143. [[PubMed: 9445291](#)]

24. Black J, Grover B. A hazard of pressure support ventilation. *Chest*. 1988;93:333–335. [[PubMed: 3276455](#)]

25. Murphy D, Dobb G. Effect of pressure support of spontaneous breathing during intermittent mandatory ventilation. *Crit Care Med*. 1987;15:612–613. [[PubMed: 3568732](#)]

26. Jounieaux V, Duran A, Levi-Valensi P. Synchronized intermittent mandatory ventilation with and without pressure support ventilation in weaning patients with COPD from mechanical ventilation. *Chest*. 1994;105:1204–1210. [[PubMed: 8162750](#)]

27. Giuliani R, Mascia L, Recchia F, et al. Patient-ventilator interaction during synchronized intermittent mandatory ventilation. Effects of flow triggering. *Am J Respir Crit Care Med*. 1995;151:1–9. [[PubMed: 7812537](#)]
-
28. Leung P, Jubran A, Tobin MJ. Comparison of assisted ventilator modes on triggering, patient effort, and dyspnea. *Am J Respir Crit Care Med*. 1997;155:1940–1948. [[PubMed: 9196100](#)]
-
29. Marini JJ. Weaning from mechanical ventilation. *N Engl J Med*. 1991;324:1496–1497. [[PubMed: 2023609](#)]
-
30. Lofaso F, Aslanian P, Richard JC, et al. Expiratory valves used for home devices: experimental and clinical comparison. *Eur Respir J*. 1998;11:1382–1388. [[PubMed: 9657583](#)]
-
31. Lofaso F, Brochard L, Hang T, et al. Evaluation of carbon dioxide rebreathing during pressure support with BiPAP devices. *Chest*. 1995;108:772–778. [[PubMed: 7656632](#)]
-
32. Mancebo J, Amaro P, Mollo JL, et al. Comparison of the effects of pressure support ventilation delivered by three different ventilators during weaning from mechanical ventilation. *Intensive Care Med*. 1995;21:913–919. [[PubMed: 8636523](#)]
-
33. Conti G, Cinnella G, Barboni E, et al. Estimation of occlusion pressure during assisted ventilation in patients with intrinsic PEEP. *Am J Respir Crit Care Med*. 1996;154:907–912. [[PubMed: 8887584](#)]
-
34. Lessard MR, Brochard L. Assisted ventilation during weaning. *Curr Opin Anaesthesiol*. 1991;4:266–271.
-
35. Crimi C, Noto A, Princi P, et al. A European survey of noninvasive ventilation practices. *Eur Respir J*. 2010;36:362–369. [[PubMed: 20075052](#)]
-
36. Demoule A, Girou E, Richard JC, et al. Increased use of noninvasive ventilation in French intensive care units. *Intensive Care Med*. 2006;32:1747–1755. [[PubMed: 16799775](#)]
-
37. Schettino G, Altobelli N, Kacmarek RM. Noninvasive positive-pressure ventilation in acute respiratory failure outside clinical trials: experience at the Massachusetts General Hospital. *Crit Care Med*. 2008;36:441–447. [[PubMed: 18091540](#)]
-
38. Vitacca M, Barbano L, D’Anna S, et al. Comparison of five bilevel pressure ventilators in patients with chronic ventilatory failure: a physiologic study. *Chest*. 2002;122:2105–2114. [[PubMed: 12475854](#)]
-
39. Ferreira JC, Chipman DW, Hill NS, Kacmarek RM. Bilevel vs ICU ventilators providing noninvasive ventilation: effect of system leaks: a COPD lung model comparison. *Chest*. 2009;136:448–456. [[PubMed: 19429723](#)]
-
40. Vignaux L, Tassaux D, Jolliet P. Performance of noninvasive ventilation modes on ICU ventilators during pressure support: a bench model study. *Intensive Care Med*. 2007;33:1444–1451. [[PubMed: 17563875](#)]
-
41. Vignaux L, Tassaux D, Carteaux G, et al. Performance of noninvasive ventilation algorithms on ICU ventilators during pressure support: a clinical study. *Intensive Care Med*. 2010;36:2053–2059. [[PubMed: 20689921](#)]
-
42. Carteaux G, Lyazidi A, Thille AW, Brochard L. Comparison of patient ventilator asynchrony using ICU ventilators and a NIV ventilator (BiPAP Vision, Respironics): preliminary results. *Intensive Care Med*. 2009;35:S22.
-

43. Brochard L, Harf A, Lorino H, Lemaire F. Inspiratory pressure support prevents diaphragmatic fatigue during weaning from mechanical ventilation. *Am Rev Respir Dis.* 1989;139:513–521. [[PubMed: 2643905](#)]
-
44. Ershowsky P, Krieger B. Changes in breathing pattern during pressure support ventilation. *Respir Care.* 1987;32:1011–1016.
-
45. Tokioka H, Saito S, Kosaka F. Effect of pressure support ventilation on breathing pattern and respiratory work. *Intensive Care Med.* 1989;15:491–494. [[PubMed: 2607036](#)]
-
46. Van de Graaff WB, Gordey K, Dornseif SE, et al. Pressure support. Changes in ventilatory pattern and components of the work of breathing. *Chest.* 1991;100:1082–1089.
-
47. Hurst J, Branson R, Davis K, Barrette R. Cardiopulmonary effects of pressure support ventilation. *Arch Surg.* 1989;124:1067–1070. [[PubMed: 2774910](#)]
-
48. MacIntyre N. Pressure support ventilation: effects on ventilatory reflexes and ventilatory muscle work load. *Respir Care.* 1987;32:447–457.
-
49. Tobin MJ, Perez W, Guenther SM, et al. The pattern of breathing during successful and unsuccessful trials of weaning from mechanical ventilation. *Am Rev Respir Dis.* 1986;134:1111–1118. [[PubMed: 3789513](#)]
-
50. Viale JP, Duperret S, Mahul P, et al. Time course evolution of ventilatory responses to inspiratory unloading in patients. *Am J Respir Crit Care Med.* 1998;157:428–434. [[PubMed: 9476854](#)]
-
51. El-Khatib M, Bou-Khalil P, Zeineldine S, et al. Metabolic and respiratory variables during pressure support versus synchronized intermittent mandatory ventilation. *Respiration.* 2009;77:154–159. [[PubMed: 18547941](#)]
-
52. Jamaledine G, El-Khatib M. Pressure support ventilation: a significant confounding factor in the determination of the rapid shallow breathing index. *Crit Care Med.* 2008;36:1692–1693; author reply 1693–1694. [[PubMed: 18448953](#)]
-
53. Lofaso F, Isabey D, Lorino H, et al. Respiratory response to positive and negative inspiratory pressure in humans. *Respir Physiol.* 1992;89:75–88. [[PubMed: 1518988](#)]
-
54. Giannouli E, Webster K, Roberts D, Younes M. Response of ventilator-dependent patients to different levels of pressure support and proportional assist. *Am J Respir Crit Care Med.* 1999;159:1716–1725. [[PubMed: 10351909](#)]
-
55. Tassaux D, Michotte JB, Gainnier M, et al. Expiratory trigger setting in pressure support ventilation: from mathematical model to bedside. *Crit Care Med.* 2004;32:1844–1850. [[PubMed: 15343011](#)]
-
56. Beck J, Gottfried SB, Navalesi P, et al. Electrical activity of the diaphragm during pressure support ventilation in acute respiratory failure. *Am J Respir Crit Care Med.* 2001;164:419–424. [[PubMed: 11500343](#)]
-
57. Pesenti A, Rossi N, Calori A, et al. Effects of short-term oxygenation changes on acute lung injury patients undergoing pressure support ventilation. *Chest.* 1993;103:1185–1189. [[PubMed: 8131462](#)]
-

58. Volta CA, Alvisi V, Bertacchini S, et al. Acute effects of hyperoxemia on dyspnoea and respiratory variables during pressure support ventilation. *Intensive Care Med.* 2006;32:223–229. [[PubMed: 16432678](#)]
-
59. Bonmarchand G, Chevron V, Chopin C, et al. Increased initial flow rate reduces inspiratory work of breathing during pressure support ventilation in patients with exacerbation of chronic obstructive pulmonary disease. *Intensive Care Med.* 1996;22:1147–1154. [[PubMed: 9120105](#)]
-
60. Lofaso F, Isabey D, Harf A, Scheid P. Airway anesthesia during positive and negative inspiratory pressure breathing in man. *Respir Physiol.* 1992;89:89–96. [[PubMed: 1518989](#)]
-
61. MacIntyre NR, Leatherman N. Ventilatory muscle load and the frequency-tidal volume pattern during inspiratory pressure-assisted (pressure supported) ventilation. *Am Rev Respir Dis.* 1990;141:327–331. [[PubMed: 2405758](#)]
-
62. Parthasarathy S, Tobin MJ. Effect of ventilator mode on sleep quality in critically ill patients. *Am J Respir Crit Care Med.* 2002;166:1423–1429. [[PubMed: 12406837](#)]
-
63. Valentine DD, Hammond MD, Downs JB, et al. Distribution of ventilation and perfusion with different modes of mechanical ventilation. *Am Rev Respir Dis.* 1991;143:1262–1266. [[PubMed: 2048810](#)]
-
64. Beydon L, Cinotti L, Rekik N, et al. Changes in the distribution of ventilation and perfusion associated with separation from mechanical ventilation in patients with obstructive pulmonary disease. *Anesthesiology.* 1991;75:730–738. [[PubMed: 1952197](#)]
-
65. Diaz O, Iglesia R, Ferrer M, et al. Effects of noninvasive ventilation on pulmonary gas exchange and hemodynamics during acute hypercapnic exacerbations of chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 1997;156:1840–1845. [[PubMed: 9412564](#)]
-
66. Ferrer M, Iglesia R, Roca J, et al. Pulmonary gas exchange response to weaning with pressure-support ventilation in exacerbated chronic obstructive pulmonary disease patients. *Intensive Care Med.* 2002;28:1595–1599. [[PubMed: 12415446](#)]
-
67. Putensen C, Mutz NJ, Putensen-Himmer G, Zinserling J. Spontaneous breathing during ventilatory support improves ventilation-perfusion distribution in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med.* 1999;159:1241–1248. [[PubMed: 10194172](#)]
-
68. Putensen C, Rasanen J, Lopez FA. Interfacing between spontaneous breathing and mechanical ventilation affects ventilation-perfusion distributions in experimental bronchoconstriction. *Am J Respir Crit Care Med.* 1995;151:993–999. [[PubMed: 7697278](#)]
-
69. Pinto Da Costa N, Di Marco F, Lyazidi A, et al. Effect of pressure support on end-expiratory lung volume and lung diffusion for carbon monoxide. *Crit Care Med.* 2011;39:2283–2289.
-
70. Zeravik J, Borg U, Pfeiffer U. Efficacy of pressure support ventilation dependent on extravascular lung water. *Chest.* 1990;97:1412–1419. [[PubMed: 2189696](#)]
-

71. Annat GJ, Viale JP, Derymez CP, et al. [Oxygen](#) cost of breathing and diaphragmatic pressure-time index. Measurement in patients with COPD during weaning with pressure support ventilation. *Chest*. 1990;98:411–414. [\[PubMed: 2198141\]](#)
-
72. Viale JP, Annat GJ, Bouffard YM, et al. [Oxygen](#) cost of breathing in post-operative patients: pressure support ventilation vs continuous positive airway pressure. *Chest*. 1988;93:506–509. [\[PubMed: 3125013\]](#)
-
73. Hilbert G, Choukroun ML, Benissan GG, et al. Optimal pressure support level for beginning weaning in patients with COPD: measurement of diaphragmatic activity with step-by-step decreasing pressure support level. *J Crit Care*. 1998;13:110–118. [\[PubMed: 9758025\]](#)
-
74. Jubran A, Van de Graaff WB, Tobin MJ. Variability of patient-ventilator interaction with pressure support ventilation in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med*. 1995;152:129–136. [\[PubMed: 7599811\]](#)
-
75. Fauroux B, Hart N, Luo YM, et al. Measurement of diaphragm loading during pressure support ventilation. *Intensive Care Med*. 2003;29:1960–1966. [\[PubMed: 14556049\]](#)
-
76. Brochard L. Pressure support ventilation. In: Marini JJ, Roussos C, eds. *Ventilatory Failure*. Berlin, Germany: Springer-Verlag; 1991:381–391.
-
77. Brochard L, Rua F, Lorino H, et al. Inspiratory pressure support compensates for the additional work of breathing caused by the endotracheal tube. *Anesthesiology*. 1991;75:739–745. [\[PubMed: 1952198\]](#)
-
78. Shapiro M, Wilson RK, Casar G, et al. Work of breathing through different sized endotracheal tubes. *Crit Care Med*. 1986;14:1028–1031. [\[PubMed: 3780244\]](#)
-
79. Fiastro JF, Habib MP, Quan SF. Pressure support compensation for inspiratory work due to endotracheal tubes and demand continuous positive airway pressure. *Chest*. 1988;93:499–505. [\[PubMed: 3277803\]](#)
-
80. Nathan SD, Ishaaya AM, Koerner SK, Belman MJ. Prediction of minimal pressure support during weaning from mechanical ventilation. *Chest*. 1993;103:1215–1219. [\[PubMed: 8131468\]](#)
-
81. Ishaaya AM, Nathan SD, Belman MJ. Work of breathing after extubation. *Chest*. 1995;107:204–209. [\[PubMed: 7813279\]](#)
-
82. Straus C, Louis B, Isabey D, et al. Contribution of the endotracheal tube and the upper airway to breathing workload. *Am J Respir Crit Care Med*. 1998;157:23–30. [\[PubMed: 9445274\]](#)
-
83. Sassoon CSH, Light RW, Lodia R, et al. Pressure-time product during continuous positive airway pressure, pressure support ventilation and T-piece during weaning from mechanical ventilation. *Am Rev Respir Dis*. 1991;143:459–475.
-
84. Sassoon CSH, Lodia R, Rheeman CH, et al. Inspiratory muscle work of breathing during flow-by, demand-flow, and continuous-flow systems in patients with chronic obstructive pulmonary disease. *Am Rev Respir Dis*. 1992;145:1219–1222. [\[PubMed: 1586070\]](#)
-

85. Esteban A, Alia I, Gordo F, et al. Extubation outcome after spontaneous breathing trials with T-tube or pressure support ventilation. *Am J Respir Crit Care Med*. 1997;156:459–465. [[PubMed: 9279224](#)]
-
86. Farias J-A, Retta A, Alia I, et al. A comparison of two methods to perform a breathing trial before extubation in pediatric intensive care patients. *Intensive Care Med*. 2001;27:1649–1654. [[PubMed: 11685307](#)]
-
87. Matic I, Danic D, Majeric-Kogler V, et al. Chronic obstructive pulmonary disease and weaning of difficult-to-wean patients from mechanical ventilation: randomized prospective study. *Croat Med J*. 2007;48:51–58. [[PubMed: 17309139](#)]
-
88. Ezingear E, Diconne E, Guyomarc'h S, et al. Weaning from mechanical ventilation with pressure support in patients failing a T-tube trial of spontaneous breathing. *Intensive Care Med*. 2006;32:165–169. [[PubMed: 16283162](#)]
-
89. Cabello B, Thille AW, Roche-Campo F, et al. Physiological comparison of three spontaneous breathing trials in difficult-to-wean patients. *Intensive Care Med*. 2010;36:1171–1179. [[PubMed: 20352189](#)]
-
90. Iotti GA, Olivei MC, Palo A, et al. Unfavorable mechanical effects of heat and moisture exchangers in ventilated patients. *Intensive Care Med*. 1997;23:399–405. [[PubMed: 9142578](#)]
-
91. Le Bourdelles G, Mier L, Fiquet B, et al. Comparison of the effects of heat and moisture exchangers and heated humidifiers on ventilation and gas exchange during weaning trials from mechanical ventilation. *Chest*. 1996;110:1294–1298.
-
92. Pelosi P, Solca M, Ravagnan I, et al. Effects of heat and moisture exchangers on minute ventilation, ventilatory drive, and work of breathing during pressure-support ventilation in acute respiratory failure. *Crit Care Med*. 1996;24:1184–1188. [[PubMed: 8674333](#)]
-
93. Campbell RS, Davis K Jr, Johannigman JA, Branson RD. The effects of passive humidifier dead space on respiratory variables in paralyzed and spontaneously breathing patients. *Respir Care*. 2000;45:306–312. [[PubMed: 10771799](#)]
-
94. Girault C, Breton L, Richard JC, et al. Mechanical effects of airway humidification devices in difficult to wean patients. *Crit Care Med*. 2003;31:1306–1311. [[PubMed: 12771595](#)]
-
95. Natalini G, Bardini P, Latronico N, Candiani A. Impact of heat and moisture exchangers on ventilatory pattern and respiratory mechanics in spontaneously breathing patients. *Monaldi Arch Chest Dis*. 1994;49:561–564. [[PubMed: 7711718](#)]
-
96. Jaber S, Chanques G, Matecki S, et al. Comparison of the effects of heat and moisture exchangers and heated humidifiers on ventilation and gas exchange during non-invasive ventilation. *Intensive Care Med*. 2002;28:1590–1594. [[PubMed: 12415445](#)]
-
97. Lellouche F, Maggiore SM, Deye N, et al. Effect of the humidification device on the work of breathing during noninvasive ventilation. *Intensive Care Med*. 2002;28:1582–1589. [[PubMed: 12415444](#)]
-
98. Boyer A, Vargas F, Hilbert G, et al. Small dead space heat and moisture exchangers do not impede gas exchange during noninvasive ventilation: a comparison with a heated humidifier. *Intensive Care Med*. 2010;36:1348–1354. [[PubMed: 20422150](#)]

-
99. Drouot X, Cabello B, d'Ortho MP, Brochard L. Sleep in the intensive care unit. *Sleep Med Rev.* 2008;12:391–403. [[PubMed: 18502155](#)]
-
100. Parthasarathy S, Tobin MJ. Sleep in the intensive care unit. *Intensive Care Med.* 2004;30:197–206. [[PubMed: 14564378](#)]
-
101. Bosma K, Ferreyra G, Ambrogio C, et al. Patient-ventilator interaction and sleep in mechanically ventilated patients: pressure support versus proportional assist ventilation. *Crit Care Med.* 2007;35:1048–1054. [[PubMed: 17334259](#)]
-
102. Cabello B, Thille AW, Drouot X, et al. Sleep quality in mechanically ventilated patients: comparison of three ventilatory modes. *Crit Care Med.* 2008;36:1749–1755. [[PubMed: 18496373](#)]
-
103. Fanfulla F, Delmastro M, Berardinelli A, et al. Effects of different ventilator settings on sleep and inspiratory effort in patients with neuromuscular disease. *Am J Respir Crit Care Med.* 2005;172:619–624. [[PubMed: 15961699](#)]
-
104. Dojat M, Brochard L, Lemaire F, Harf A. A knowledge based systems for assisted ventilation of patients in intensive care units. *Int J Clin Monit Comput.* 1992;9:239–250. [[PubMed: 1484275](#)]
-
105. Thille AW, Rodriguez P, Cabello B, et al. Patient-ventilator asynchrony during assisted mechanical ventilation. *Intensive Care Med.* 2006;32:1515–1522. [[PubMed: 16896854](#)]
-
106. Fabry B, Guttman J, Eberhard L, et al. An analysis of desynchronization between the spontaneously breathing patient and ventilator during inspiratory pressure support. *Chest.* 1995;107:1387–1394. [[PubMed: 7750336](#)]
-
107. Nava S, Bruschi C, Fracchia C, et al. Patient-ventilator interaction and inspiratory effort during pressure support ventilation in patients with different pathologies. *Eur Respir J.* 1997;10:177–183. [[PubMed: 9032512](#)]
-
108. Nava S, Bruschi C, Rubini F, et al. Respiratory response and inspiratory effort during pressure support ventilation in COPD patients. *Intensive Care Med.* 1995;21:871–879. [[PubMed: 8636518](#)]
-
109. Chao DC, Scheinhorn DJ, Stearn-Hassenpflug M. Patient-ventilator trigger asynchrony in prolonged mechanical ventilation. *Chest.* 1997;112:1592–1599. [[PubMed: 9404759](#)]
-
110. Nava S, Ambrosino N, Bruschi C, et al. Physiological effects of flow and pressure triggering during non-invasive mechanical ventilation in patients with chronic obstructive pulmonary disease. *Thorax.* 1997;52:249–254. [[PubMed: 9093341](#)]
-
111. Ambrosino N, Foglio K, Rubini F, et al. Non-invasive mechanical ventilation in acute respiratory failure due to chronic obstructive pulmonary disease: correlates for success. *Thorax.* 1995;50:755–757. [[PubMed: 7570410](#)]
-
112. Vitacca M, Bianchi L, Zanotti E, et al. Assessment of physiologic variables and subjective comfort under different levels of pressure support ventilation. *Chest.* 2004;126:851–859. [[PubMed: 15364766](#)]
-
113. Calderini E, Confalonieri M, Puccio PG, et al. Patient-ventilator asynchrony during noninvasive ventilation: the role of expiratory trigger. *Intensive Care Med.* 1999;25:662–667. [[PubMed: 10470568](#)]
-

114. de Wit M, Miller KB, Green DA, et al. Ineffective triggering predicts increased duration of mechanical ventilation. *Crit Care Med*. 2009;37:2740–2745.
-
115. Thille AW, Cabello B, Galia F, et al. Reduction of patient-ventilator asynchrony by reducing tidal volume during pressure-support ventilation. *Intensive Care Med*. 2008;34:1477–1486. [[PubMed: 18437356](#)]
-
116. Georgopoulos D, Prinianakis G, Kondili E. Bedside waveforms interpretation as a tool to identify patient-ventilator asynchronies. *Intensive Care Med*. 2006;32:34–47. [[PubMed: 16283171](#)]
-
117. Sassoos CS, Lodia R, Rheeman CH, et al. Inspiratory muscle work of breathing during flow-by, demand-flow, and continuous-flow systems in patients with chronic obstructive pulmonary disease. *Am Rev Respir Dis*. 1992;145:1219–1222. [[PubMed: 1586070](#)]
-
118. Branson RD, Campbell RS, Davis K Jr, Johnson DJ 2nd. Comparison of pressure and flow triggering systems during continuous positive airway pressure. *Chest*. 1994;106:540–544. [[PubMed: 7774334](#)]
-
119. Sassoos CS, Gruer SE. Characteristics of the ventilator pressure- and flow-trigger variables. *Intensive Care Med*. 1995;21:159–168. [[PubMed: 7775698](#)]
-
120. Ranieri VM, Mascia L, Petruzzelli V, et al. Inspiratory effort and measurement of dynamic intrinsic PEEP in COPD patients: effects of ventilator triggering systems. *Intensive Care Med*. 1995;21:896–903. [[PubMed: 8636521](#)]
-
121. Navalesi P, Costa R, Ceriana P, et al. Non-invasive ventilation in chronic obstructive pulmonary disease patients: helmet versus facial mask. *Intensive Care Med*. 2007;33:74–81. [[PubMed: 17039354](#)]
-
122. Vargas F, Thille A, Lyazidi A, et al. Helmet with specific settings versus facemask for noninvasive ventilation. *Crit Care Med*. 2009;37:1921–1928. [[PubMed: 19384209](#)]
-
123. Appendini L, Purro A, Patessio A, et al. Partitioning of inspiratory muscle workload and pressure assistance in ventilator-dependent COPD patients. *Am J Respir Crit Care Med*. 1996;154:1301–1309. [[PubMed: 8912740](#)]
-
124. Vitacca M, Lanini B, Nava S, et al. Inspiratory muscle workload due to dynamic intrinsic PEEP in stable COPD patients: effects of two different settings of non-invasive pressure-support ventilation. *Monaldi Arch Chest Dis*. 2004;61:81–85. [[PubMed: 15510707](#)]
-
125. Tassaux D, Gainnier M, Battisti A, Joliet P. Impact of expiratory trigger setting on delayed cycling and inspiratory muscle workload. *Am J Respir Crit Care Med*. 2005;172:1283–1289. [[PubMed: 16109983](#)]
-
126. Imanaka H, Nishimura M, Takeuchi M, et al. Autotriggering caused by cardiogenic oscillation during flow-triggered mechanical ventilation. *Crit Care Med*. 2000;28:402–407. [[PubMed: 10708174](#)]
-
127. Prinianakis G, Kondili E, Georgopoulos D. Effects of the flow waveform method of triggering and cycling on patient-ventilator interaction during pressure support. *Intensive Care Med*. 2003;29:1950–1959. [[PubMed: 12665996](#)]
-
128. Jubran A. Inspiratory flow rate: more may not be better. *Crit Care Med*. 1999;27:670–671. [[PubMed: 10321642](#)]
-

129. Bonmarchand G, Chevron V, Menard JF, et al. Effects of pressure ramp slope values on the work of breathing during pressure support ventilation in restrictive patients. *Crit Care Med*. 1999;27:715–722. [[PubMed: 10321660](#)]
-
130. Tokioka H, Tanaka T, Ishizu T, et al. The effect of breath termination criterion on breathing patterns and the work of breathing during pressure support ventilation. *Anesth Analg*. 2001;92:161–165. [[PubMed: 11133620](#)]
-
131. MacIntyre NR, Ho LI. Effects of initial flow rate and breath termination criteria on pressure support ventilation. *Chest*. 1991;99:134–138. [[PubMed: 1984943](#)]
-
132. MacIntyre NR, McConnell R, Cheng KC, Sane A. Patient-ventilator flow dyssynchrony: flow-limited versus pressure-limited breaths. *Crit Care Med*. 1997;25:1671–1677. [[PubMed: 9377881](#)]
-
133. Chiumello D, Pelosi P, Taccone P, et al. Effect of different inspiratory rise time and cycling off criteria during pressure support ventilation in patients recovering from acute lung injury. *Crit Care Med*. 2003;31:2604–2610. [[PubMed: 14605531](#)]
-
134. Chiumello D, Pelosi P, Croci M, et al. The effects of pressurization rate on breathing pattern, work of breathing, gas exchange and patient comfort in pressure support ventilation. *Eur Respir J*. 2001;18:107–114. [[PubMed: 11510780](#)]
-
135. Prinianakis G, Delmastro M, Carlucci A, et al. Effect of varying the pressurisation rate during noninvasive pressure support ventilation. *Eur Respir J*. 2004;23:314–320. [[PubMed: 14979510](#)]
-
136. Corne S, Gillespie D, Roberts D, Younes M. Effect of inspiratory flow rate on respiratory rate in intubated ventilated patients. *Am J Respir Crit Care Med*. 1997;156:304–308. [[PubMed: 9230766](#)]
-
137. Laghi F, Segal J, Choe WK, Tobin MJ. Effect of imposed inflation time on respiratory frequency and hyperinflation in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med*. 2001;163:1365–1370. [[PubMed: 11371402](#)]
-
138. Parthasarathy S, Jubran A, Tobin MJ. Cycling of inspiratory and expiratory muscle groups with the ventilator in airflow limitation. *Am J Respir Crit Care Med*. 1998;158:1471–1478. [[PubMed: 9817695](#)]
-
139. Spahija J, de Marchie M, Albert M, et al. Patient-ventilator interaction during pressure support ventilation and neurally adjusted ventilatory assist. *Crit Care Med*. 2010;38:518–526. [[PubMed: 20083921](#)]
-
140. Younes M. Proportional assist ventilation and pressure support ventilation; similarities and differences. *Update Intensive Care Emerg Med*. 1991;15:361–380.
-
141. Yamada Y, Du HL. Analysis of the mechanisms of expiratory asynchrony in pressure support ventilation: a mathematical approach. *J Appl Physiol*. 2000;88:2143–2150. [[PubMed: 10846029](#)]
-
142. Younes M, Kun J, Webster K, Roberts D. Response of ventilator-dependent patients to delayed opening of exhalation valve. *Am J Respir Crit Care Med*. 2002;166:21–30. [[PubMed: 12091166](#)]
-
143. Lessard MR, Lofaso F, Brochard L. Expiratory muscle activity increases intrinsic positive end-expiratory pressure independently of dynamic hyperinflation in mechanically ventilated patients. *Am J Respir Crit Care Med*. 1995;151:562–

569. [\[PubMed: 7842221\]](#)

144. Prinianakis G, Plataki M, Kondili E, et al. Effects of relaxation of inspiratory muscles on ventilator pressure during pressure support. *Intensive Care Med.* 2008;34:70–74. [\[PubMed: 17928993\]](#)

145. Vignaux L, Vargas F, Roeseler J, et al. Patient-ventilator asynchrony during non-invasive ventilation for acute respiratory failure: a multicenter study. *Intensive Care Med.* 2009;35:840–846. [\[PubMed: 19183949\]](#)

146. Antonelli M, Conti G, Pelosi P, et al. New treatment of acute hypoxemic respiratory failure: noninvasive pressure support ventilation delivered by helmet—a pilot controlled trial. *Crit Care Med.* 2002;30:602–608. [\[PubMed: 11990923\]](#)

147. Hotchkiss JR Jr, Adams AB, Stone MK, et al. Oscillations and noise: inherent instability of pressure support ventilation? *Am J Respir Crit Care Med.* 2002;165:47–53. [\[PubMed: 11779729\]](#)

148. Branson R. Understanding and implementing advances in ventilator capabilities. *Curr Opin Crit Care.* 2004;10:23–32. [\[PubMed: 15166846\]](#)

149. Du HL, Amato MB, Yamada Y. Automation of expiratory trigger sensitivity in pressure support ventilation. *Respir Care Clin N Am.* 2001;7:503–517, x. [\[PubMed: 11517036\]](#)

150. Du HL, Ohtsuiji M, Shigeta M, et al. Expiratory asynchrony in proportional assist ventilation. *Am J Respir Crit Care Med.* 2002;165:972–977. [\[PubMed: 11934724\]](#)

151. Kamat S, Dulfano M, Segal M. The effects of intermittent positive pressure breathing (IPPB/I) with compressed air in patients with severe chronic nonspecific obstructive pulmonary disease. *Am Rev Respir Dis.* 1932;86:360–380.

152. Ayres S, Kozam R, Lukas D. The effects of intermittent positive pressure breathing on intrathoracic pressure, pulmonary mechanics, and the work of breathing. *Am Rev Respir Dis.* 1963;87:370–379. [\[PubMed: 13968920\]](#)

153. Sukumalchantra Y, Park S, Williams MJ. The effects of intermittent positive pressure breathing (IPPB) in acute ventilatory failure. *Am Rev Respir Dis.* 1965;92:885–893. [\[PubMed: 5320894\]](#)

154. Jones R, McNamara J, Gaensler E. The effects of intermittent positive pressure breathing on simulated pulmonary obstruction. *Am Rev Respir Dis.* 1960;92:885–893.

155. Mancebo J, Isabey D, Lorino H, et al. Comparative effects of pressure support ventilation and intermittent positive pressure breathing (IPPB) in non-intubated healthy subjects. *Eur Respir J.* 1995;8:1901–1909. [\[PubMed: 8620960\]](#)

156. Dreyfuss D, Saumon G. Ventilator-induced lung injury: lessons from experimental studies. *Am J Respir Crit Care Med.* 1998;157:294–323. [\[PubMed: 9445314\]](#)

157. Dreyfuss D, Saumon G. From ventilator-induced lung injury to multiple organ dysfunction? *Intensive Care Med.* 1998;24:102–104. [\[PubMed: 9539065\]](#)

158. Foti G, Cereda M, Banfi G, et al. End-inspiratory airway occlusion: a method to assess the pressure developed by inspiratory muscles in patients with acute lung injury undergoing pressure support. *Am J Respir Crit Care Med.*

1997;156:1210–1216. [\[PubMed: 9351624\]](#)

159. Cinnella G, Conti G, Lofaso F, et al. Effects of assisted ventilation on the work of breathing: volume-controlled versus pressure-controlled ventilation. *Am J Respir Crit Care Med*. 1996;153:1025–1033. [\[PubMed: 8630541\]](#)

160. Marini JJ, Smith TC, Lamb VT. External work output and force generation during synchronized intermittent mechanical ventilation. *Am Rev Respir Dis*. 1988;138:1169–1179. [\[PubMed: 3202477\]](#)

161. Imsand C, Feihl F, Perret C, Fitting JW. Regulation of inspiratory neuromuscular output during synchronized intermittent mechanical ventilation. *Anesthesiology*. 1994;80:13–22. [\[PubMed: 8291702\]](#)

162. Knebel AR, Janson-Bjerklie SL, Malley JD, et al. Comparison of breathing comfort during weaning with two ventilatory modes. *Am J Respir Crit Care Med*. 1994;149:14–18. [\[PubMed: 8111572\]](#)

163. Uchiyama A, Imanaka H, Taenaka N, et al. Comparative evaluation of diaphragmatic activity during pressure support ventilation and intermittent mandatory ventilation in animal model. *Am J Respir Crit Care Med*. 1994;150:1564–1568. [\[PubMed: 7952615\]](#)

164. Tobin MJ, Jubran A, Laghi F. Patient-ventilator interaction. *Am J Respir Crit Care Med*. 2001;163:1059–1063. [\[PubMed: 11316635\]](#)

165. Younes M. Proportional assist ventilation, a new approach to ventilatory support. Theory. *Am Rev Respir Dis*. 1992;145:114–120. [\[PubMed: 1731573\]](#)

166. Younes M, Puddy A, Roberts D, et al. Proportional assist ventilation. Results of an initial clinical trial. *Am Rev Respir Dis*. 1992;145:121–129. [\[PubMed: 1731574\]](#)

167. Gay PC, Hess DR, Hill NS. Noninvasive proportional assist ventilation for acute respiratory insufficiency. Comparison with pressure support ventilation. *Am J Respir Crit Care Med*. 2001;164:1606–1611. [\[PubMed: 11719297\]](#)

168. Fernandez-Vivas M, Caturla-Such J, Gonzalez de la Rosa J, et al. Noninvasive pressure support versus proportional assist ventilation in acute respiratory failure. *Intensive Care Med*. 2003;29:1126–1133. [\[PubMed: 12802487\]](#)

169. Ambrosino N, Rossi A. Proportional assist ventilation (PAV): a significant advance or a futile struggle between logic and practice? *Thorax*. 2002;57:272–276. [\[PubMed: 11867835\]](#)

170. Vitacca M. New things are not always better: proportional assist ventilation vs. pressure support ventilation. *Intensive Care Med*. 2003;29:1038–1040. [\[PubMed: 12918515\]](#)

171. Branson RD, Johannigman JA. What is the evidence base for the newer ventilation modes? *Respir Care*. 2004;49:742–760. [\[PubMed: 15222907\]](#)

172. Younes M. Proportional assist ventilation. In: Tobin MJ, ed. *Principles and Practice of Mechanical Ventilation*. New York, NY: McGraw-Hill; 1994:349–369.

173. Ranieri VM, Giuliani R, Mascia L, et al. Patient-ventilator interaction during acute hypercapnia: pressure-support vs. proportional-assist ventilation. *J Appl Physiol*. 1996;81:426–436. [[PubMed: 8828695](#)]
-
174. Grasso S, Puntillo F, Mascia L, et al. Compensation for increase in respiratory workload during mechanical ventilation. Pressure-support versus proportional-assist ventilation. *Am J Respir Crit Care Med*. 2000;161:819–826. [[PubMed: 10712328](#)]
-
175. Wysocki M, Meshaka P, Richard JC, Similowski T. Proportional-assist ventilation compared with pressure-support ventilation during exercise in volunteers with external thoracic restriction. *Crit Care Med*. 2004;32:409–414. [[PubMed: 14758156](#)]
-
176. Mols G, von Ungern-Sternberg B, Rohr E, Haberthur C, et al. Respiratory comfort and breathing pattern during volume proportional assist ventilation and pressure support ventilation: a study on volunteers with artificially reduced compliance. *Crit Care Med*. 2000;28:1940–1946. [[PubMed: 10890645](#)]
-
177. Navalesi P, Hernandez P, Wongs A, et al. Proportional assist ventilation in acute respiratory failure: effects on breathing pattern and inspiratory effort. *Am J Respir Crit Care Med*. 1996;154:1330–1338. [[PubMed: 8912744](#)]
-
178. Passam F, Hoing S, Prinianakis G, et al. Effect of different levels of pressure support and proportional assist ventilation on breathing pattern, work of breathing and gas exchange in mechanically ventilated hypercapnic COPD patients with acute respiratory failure. *Respiration*. 2003;70:355–361. [[PubMed: 14512669](#)]
-
179. Purro A, Appendini L, De Gaetano A, et al. Physiologic determinants of ventilator dependence in long-term mechanically ventilated patients. *Am J Respir Crit Care Med*. 2000;161:1115–1123. [[PubMed: 10764299](#)]
-
180. Wrigge H, Golisch W, Zinserling J, et al. Proportional assist versus pressure support ventilation: effects on breathing pattern and respiratory work of patients with chronic obstructive pulmonary disease. *Intensive Care Med*. 1999;25:790–798. [[PubMed: 10447535](#)]
-
181. Wysocki M, Richard JC, Meshaka P. Noninvasive proportional assist ventilation compared with noninvasive pressure support ventilation in hypercapnic acute respiratory failure. *Crit Care Med*. 2002;30:323–329. [[PubMed: 11889302](#)]
-
182. Porta R, Appendini L, Vitacca M, et al. Mask proportional assist vs pressure support ventilation in patients in clinically stable condition with chronic ventilatory failure. *Chest*. 2002;122:479–488. [[PubMed: 12171820](#)]
-
183. Delaere S, Roeseler J, D’Hoore W, et al. Respiratory muscle workload in intubated, spontaneously breathing patients without COPD: pressure support vs proportional assist ventilation. *Intensive Care Med*. 2003;29:949–954. [[PubMed: 12664221](#)]
-
184. Ranieri VM, Grasso S, Mascia L, et al. Effects of proportional assist ventilation on inspiratory muscle effort in patients with chronic obstructive pulmonary disease and acute respiratory failure. *Anesthesiology*. 1997;86:79–91. [[PubMed: 9009943](#)]
-
185. Ambrosino N, Vitacca M, Polese G, et al. Short-term effects of nasal proportional assist ventilation in patients with chronic hypercapnic respiratory insufficiency. *Eur Respir J*. 1997;10:2829–2834. [[PubMed: 9493669](#)]
-

186. Bianchi L, Foglio K, Pagani M, et al. Effects of proportional assist ventilation on exercise tolerance in COPD patients with chronic hypercapnia. *Eur Respir J*. 1998;11:422–427. [[PubMed: 9551748](#)]
-
187. Hernandez P, Maltais F, Gursahaney A, et al. Proportional assist ventilation may improve exercise performance in severe chronic obstructive pulmonary disease. *J Cardiopulm Rehabil*. 2001;21:135–142. [[PubMed: 11409222](#)]
-
188. Hart N, Hunt A, Polkey MI, et al. Comparison of proportional assist ventilation and pressure support ventilation in chronic respiratory failure due to neuromuscular and chest wall deformity. *Thorax*. 2002;57:979–981. [[PubMed: 12403882](#)]
-
189. Younes M, Kun J, Masiowski B, et al. A method for noninvasive determination of inspiratory resistance during proportional assist ventilation. *Am J Respir Crit Care Med*. 2001;163:829–839. [[PubMed: 11282753](#)]
-
190. Younes M, Webster K, Kun J, et al. A method for measuring passive elastance during proportional assist ventilation. *Am J Respir Crit Care Med*. 2001;164:50–60. [[PubMed: 11435238](#)]
-
191. Kondili E, Prinianakis G, Alexopoulou C, et al. Respiratory load compensation during mechanical ventilation—proportional assist ventilation with load-adjustable gain factors versus pressure support. *Intensive Care Med*. 2006;32:692–699. [[PubMed: 16523329](#)]
-
192. Xirouchaki N, Kondili E, Vaporidi K, et al. Proportional assist ventilation with load-adjustable gain factors in critically ill patients: comparison with pressure support. *Intensive Care Med*. 2008;34:2026–2034. [[PubMed: 18607562](#)]
-
193. Sinderby C, Navalesi P, Beck J, et al. Neural control of mechanical ventilation in respiratory failure. *Nat Med*. 1999;5:1433–1436. [[PubMed: 10581089](#)]
-
194. Beck KC, Offord KP, Scanlon PD. Comparison of four methods for calculating diffusing capacity by the single breath method. *Chest*. 1994;105:594–600. [[PubMed: 8306770](#)]
-
195. Spahija J, Beck J, de Marchie M, et al. Closed-loop control of respiratory drive using pressure support ventilation: target drive ventilation. *Am J Respir Crit Care Med*. 2005;171:1009–1014. [[PubMed: 15665323](#)]
-
196. Terzi N, Pelieu I, Guittet L, et al. Neurally adjusted ventilatory assist in patients recovering spontaneous breathing after acute respiratory distress syndrome: physiological evaluation. *Crit Care Med*. 2010;38:1830–1837. [[PubMed: 20639752](#)]
-
197. Schmidt M, Demoule A, Cracco C, et al. Neurally adjusted ventilatory assist increases respiratory variability and complexity in acute respiratory failure. *Anesthesiology*. 2010;112:670–681. [[PubMed: 20179505](#)]
-
198. Piquilloud L, Vignaux L, Bialais E, et al. Neurally adjusted ventilatory assist improves patient-ventilator interaction. *Intensive Care Med*. 2011;37:263–271. [[PubMed: 20871978](#)]
-
199. Fassoulaki A, Eforakopoulorr M. Cardiovascular, respiratory and metabolic changes produced by pressure-supported ventilation in intensive care patients. *Crit Care Med*. 1989;17:527–529. [[PubMed: 2656097](#)]
-

200. Wong D, Stemmer E, Gordon I. Acute massive air leak and pressure support ventilation. *Crit Care Med*. 1990;18:114–115. [[PubMed: 2293957](#)]
-
201. Brochard L, Rauss A, Benito S, et al. Comparison of three methods of gradual withdrawal from ventilatory support during weaning from mechanical ventilation. *Am J Respir Crit Care Med*. 1994;150:896–903. [[PubMed: 7921460](#)]
-
202. Esteban A, Frutos F, Tobin MJ, et al. A comparison of four methods of weaning patients from mechanical ventilation. Spanish Lung Failure Collaborative Group. *N Engl J Med*. 1995;332:345–350. [[PubMed: 7823995](#)]
-
203. MacIntyre NR. Weaning from mechanical ventilatory support: volume-assisting intermittent breaths versus pressure-assisting every breath. *Respir Care*. 1988;33:121–125.
-
204. Hörmann C, Baum M, Luz G, et al. Tidal volume, breathing frequency and oxygen consumption at different pressure support levels in the early stage of weaning in patients without chronic obstructive pulmonary disease. *Intensive Care Med*. 1992;18:226–230. [[PubMed: 23268516](#)]
-
205. Boyer F, Bruneau B, Gaussorgues P, et al. Aide inspiratoire avec asservissement du niveau de pression: volume ventilé minute versus fréquence ventilatoire. *Rean Soins Intens Med Urgence*. 1989;5.
-
206. Ranieri VM. Optimization of patient-ventilator interactions: closed-loop technology to turn the century. *Intensive Care Med*. 1997;23:936–939. [[PubMed: 9347363](#)]
-
207. Branson R, MacIntyre NR. Dual-control modes of mechanical ventilation. *Respir Care*. 1996;41:294–305.
-
208. Keenan H, Martin L. Volume support ventilation in infants and children: analysis of a case series. *Respir Care*. 1997;42:281–287.
-
209. Sottiaux T. Patient-ventilator interactions during volume-support ventilation: asynchrony and tidal volume instability—a report of three cases. *Respir Care*. 2001;46:255–262. [[PubMed: 11262552](#)]
-
210. Randolph AG, Wypij D, Venkataraman ST, et al. Effect of mechanical ventilator weaning protocols on respiratory outcomes in infants and children: a randomized controlled trial. *JAMA*. 2002;288:2561–2568. [[PubMed: 12444863](#)]
-
211. Jaber S, Delay JM, Matecki S, et al. Volume-guaranteed pressure-support ventilation facing acute changes in ventilatory demand. *Intensive Care Med*. 2005;31:1181–1188. [[PubMed: 16032424](#)]
-
212. Jaber S, Sebbane M, Verzilli D, et al. Adaptive support and pressure support ventilation behavior in response to increased ventilatory demand. *Anesthesiology*. 2009;110:620–627. [[PubMed: 19225395](#)]
-
213. Mireles-Cabodevila E, Chatburn RL. Work of breathing in adaptive pressure control continuous mandatory ventilation. *Respir Care*. 2009;54:1467–1472. [[PubMed: 19863830](#)]
-
214. Dojat M, Harf A, Touchard D, et al. Evaluation of a knowledge-based system providing ventilatory management and decision for extubation. *Am J Respir Crit Care Med*. 1996;153:997–1004. [[PubMed: 8630586](#)]
-

215. Strickland JH, Hasson JH. A computer-controlled ventilator weaning system. *Chest*. 1991;100:1096–1099. [PubMed: 1914564]
-
216. Strickland JH, Hasson JH. A computer-controlled ventilator weaning system. *Chest*. 1993;103:1220–1226. [PubMed: 8131469]
-
217. Brunner JX. Principles and history of closed-loop controlled ventilation. *Respir Care Clin N Am*. 2001;7:341–362. [PubMed: 11517027]
-
218. Dojat M, Brochard L. Knowledge-based systems for automatic ventilatory management. *Respir Care Clin N Am*. 2001;7:379–396, viii. [PubMed: 11517029]
-
219. Nemoto T, Hatzakis GE, Thorpe CW, et al. Automatic control of pressure support mechanical ventilation using fuzzy logic. *Am J Respir Crit Care Med*. 1999;160:550–556. [PubMed: 10430727]
-
220. Dojat M, Harf A, Touchard D, et al. Clinical evaluation of a computer-controlled pressure support mode. *Am J Respir Crit Care Med*. 2000;161:1161–1166. [PubMed: 10764306]
-
221. Bouadma L, Lellouche F, Cabello B, et al. Computer-driven management of prolonged mechanical ventilation and weaning: a pilot study. *Intensive Care Med*. 2005;31:1446–1450. [PubMed: 16132889]
-
222. Lellouche F, Mancebo J, Jolliet P, et al. A multicenter randomized trial of computer-driven protocolized weaning from mechanical ventilation. *Am J Respir Crit Care Med*. 2006;174:894–900. [PubMed: 16840741]
-
223. Rose L, Presneill JJ, Johnston L, Cade JF. A randomised, controlled trial of conventional versus automated weaning from mechanical ventilation using SmartCare/PS. *Intensive Care Med*. 2008;34:1788–1795. [PubMed: 18575843]
-
224. Schädler D, Engel C, Elke G, Pulletz S, Haake N, Frerichs I, Zick G, Scholz J, Weiler N. Automatic control of pressure support for ventilator weaning in surgical intensive care patients. *Am J Respir Crit Care Med*. 2012, in press.
-
225. Carvalho AR, Spieth PM, Güldner A, et al. Distribution of regional lung aeration and perfusion during conventional and noisy pressure support ventilation in experimental lung injury. *J Appl Physiol*. 2010;110:1083–1092.
-
226. Gama de Abreu M, Spieth PM, Pelosi P, et al. Noisy pressure support ventilation: a pilot study on a new assisted ventilation mode in experimental lung injury. *Crit Care Med*. 2008;36:818–827.
-
227. Spieth PM, Carvalho AR, Güldner A, et al. Effects of different levels of pressure support variability in experimental lung injury. *Anesthesiology*. 2009;110:342–350. [PubMed: 19194161]
-
228. Banner MJ, Euliano NR, Macintyre NR, et al. Ventilator advisory system employing load and tolerance strategy recommends appropriate pressure support ventilation settings: multisite validation study. *Chest*. 2008;133:697–703. [PubMed: 18198251]
-
229. Bonett S, Banner MJ, Euliano NR, et al. Pressure support ventilation advisory system provides valid recommendations for setting ventilator. *Respir Care*. 2011;56:271–277. [PubMed: 21235833]
-

230. Gluck E, Eubanks DH, Bone RC. Techniques for weaning a patient from mechanical ventilation; when to begin, what method to use, and how to predict outcome. *J Crit Illn*. 1993;8:121–129. [PubMed: 10148106]
-
231. Vitacca M, Vianello A, Colombo D, et al. Comparison of two methods for weaning patients with chronic obstructive pulmonary disease requiring mechanical ventilation for more than 15 days. *Am J Respir Crit Care Med*. 2001;164:225–230. [PubMed: 11463592]
-
232. Matic I, Majeric-Kogler V. Comparison of pressure support and T-tube weaning from mechanical ventilation: randomized prospective study. *Croat Med J*. 2004;45:162–166. [PubMed: 15103752]
-
233. Lellouche F, Brochard L. Advanced closed loops during mechanical ventilation (PAV, NAVA, ASV, SmartCare). *Best Pract Res Clin Anaesthesiol*. 2009;23:81–93. [PubMed: 19449618]
-
234. Leger P, Jennequin J, Gaussorgues P, Robert D. Acute respiratory failure in COPD patient treated with non invasive intermittent mechanical ventilation (control mode) with nasal mask. *Am Rev Respir Dis*. 1988;137:A63.
-
235. Brochard L, Isabey D, Piquet J, et al. Reversal of acute exacerbations of chronic obstructive lung disease by inspiratory assistance with a face mask. *N Engl J Med*. 1990;323:1523–1530. [PubMed: 2122253]
-
236. Antonelli M, Conti G, Rocco M, et al. A comparison of noninvasive positive-pressure ventilation and conventional mechanical ventilation in patients with acute respiratory failure. *N Engl J Med*. 1998;339:429–435. [PubMed: 9700176]
-
237. Girou E, Schortgen F, Delclaux C, et al. Association of noninvasive ventilation with nosocomial infections and survival in critically ill patients. *JAMA*. 2000;284:2361–2367. [PubMed: 11066187]
-
238. Hilbert G, Gruson D, Vargas F, et al. Noninvasive ventilation in immunosuppressed patients with pulmonary infiltrates, fever, and acute respiratory failure. *N Engl J Med*. 2001;344:481–487. [PubMed: 11172189]
-
239. Nava S, Ambrosino N, Clini E, et al. Noninvasive mechanical ventilation in the weaning of patients with respiratory failure due to chronic obstructive pulmonary disease. A randomized, controlled trial. *Ann Intern Med*. 1998;128:721–728. [PubMed: 9556465]
-
240. Mehta S, Hill NS. Noninvasive ventilation. *Am J Respir Crit Care Med*. 2001;163:540–577. [PubMed: 11179136]
-
241. Brochard L, Mancebo J, Wysocki M, et al. Noninvasive ventilation for acute exacerbations of chronic obstructive pulmonary disease. *N Engl J Med*. 1995;333:817–822. [PubMed: 7651472]
-
242. Maltais F, Reissmann H, Gottfried SB. Pressure support reduces inspiratory effort and dyspnea during exercise in chronic airflow obstruction. *Am J Respir Crit Care Med*. 1995;151:1027–1033. [PubMed: 7697226]
-
243. Carlucci A, Richard JC, Wysocki M, et al. Noninvasive versus conventional mechanical ventilation. An epidemiologic survey. *Am J Respir Crit Care Med*. 2001;163:874–880. [PubMed: 11282759]
-
244. Girault C, Richard JC, Chevron V, et al. Comparative physiologic effects of noninvasive assist-control and pressure support ventilation in acute hypercapnic respiratory failure. *Chest*. 1997;111:1639–1648. [PubMed: 9187187]
-

245. Hoo GW, Williams AJ. Noninvasive face-mask mechanical ventilation in patients with acute hypercapnic respiratory failure. *Chest*. 1993;103:1304–1305. [[PubMed: 8131500](#)]
-
246. Anton A, Guell R, Gomez J, et al. Predicting the result of noninvasive ventilation in severe acute exacerbations of patients with chronic airflow limitation. *Chest*. 2000;117:828–833. [[PubMed: 10713013](#)]
-
247. Poponick JM, Renston JP, Bennett RP, Emerman CL. Use of a ventilatory support system (BiPAP) for acute respiratory failure in the emergency department. *Chest*. 1999;116:166–171. [[PubMed: 10424521](#)]
-
248. Soo Hoo GW, Santiago S, Williams AJ. Nasal mechanical ventilation for hypercapnic respiratory failure in chronic obstructive pulmonary disease: determinants of success and failure. *Crit Care Med*. 1994;22:1253–1261.
-
249. L'Her E, Deye N, Lellouche F, et al. Physiologic effects of noninvasive ventilation during acute lung injury. *Am J Respir Crit Care Med*. 2005;172:1112–1118. [[PubMed: 16081548](#)]
-
250. Navalesi P, Fanfulla F, Frigerio P, et al. Physiologic evaluation of noninvasive mechanical ventilation delivered with three types of masks in patients with chronic hypercapnic respiratory failure. *Crit Care Med*. 2000;28:1785–1790. [[PubMed: 10890620](#)]
-
251. Fraticelli AT, Lellouche F, L'Her E, et al. Physiological effects of different interfaces during noninvasive ventilation for acute respiratory failure. *Crit Care Med*. 2009;37:939–945. [[PubMed: 19237900](#)]
-
252. Girault C, Daudenthun I, Chevron V, et al. Noninvasive ventilation as a systematic extubation and weaning technique in acute-on-chronic respiratory failure: a prospective, randomized controlled study. *Am J Respir Crit Care Med*. 1999;160:86–92. [[PubMed: 10390384](#)]
-
253. Vitacca M, Ambrosino N, Clini E, et al. Physiological response to pressure support ventilation delivered before and after extubation in patients not capable of totally spontaneous autonomous breathing. *Am J Respir Crit Care Med*. 2001;164:638–641. [[PubMed: 11520729](#)]
-
254. Ferrer M, Sellares J, Valencia M, et al. Non-invasive ventilation after extubation in hypercapnic patients with chronic respiratory disorders: randomised controlled trial. *Lancet*. 2009;374:1082–1088. [[PubMed: 19682735](#)]
-
255. Nava S, Gregoretti C, Fanfulla F, et al. Noninvasive ventilation to prevent respiratory failure after extubation in high-risk patients. *Crit Care Med*. 2005;33:2465–2470. [[PubMed: 16276167](#)]
-
256. Esteban A, Frutos-Vivar F, Ferguson ND, et al. Noninvasive positive-pressure ventilation for respiratory failure after extubation. *N Engl J Med*. 2004;350:2452–2460. [[PubMed: 15190137](#)]
-
257. Keenan SP, Powers C, McCormack DG, Block G. Noninvasive positive-pressure ventilation for postextubation respiratory distress: a randomized controlled trial. *JAMA*. 2002;287:3238–3244. [[PubMed: 12076220](#)]
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